



ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION
(OIC)
DEPARTMENT OF ELECTRICAL AND
ELECTRONIC
ENGINEERING

Course Name : POWER SYSTEM III LAB

Course No. : EEE 4632

Department : EEE

Section : B/B2

Project Report

Submitted By-

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Introduction:

Frequency stability is a critical aspect of power system security and integrity, focusing on the system's ability to maintain consistent frequency levels after major disturbances. In this project, we will study a 39-bus power system with 34 lines, 19 loads, 12 transformers, and 10 generators to analyze its frequency stability in the face of generation loss events. Accurate frequency control ensures that network frequency remains within a tight range around its nominal value (such as 50 Hz in many countries, including Bangladesh), preventing widespread system failures. When a significant disturbance occurs, such as the tripping of a large generator or an interconnection, the system experiences a rapid frequency decline. Synchronous generators, with their substantial mechanical inertia, provide natural damping by releasing kinetic energy, giving time for corrective actions like governor response to arrest the frequency drop. However, if the governor response is inadequate, Under Frequency Load Shedding (UFLS) mechanisms are employed to shed non-essential loads, re-establishing power balance and stabilizing the system. This project aims to evaluate the frequency excursions following a generation loss and examine the efficacy of load shedding schemes, underscoring their importance in maintaining system stability and preventing catastrophic failures. Through theoretical discussions and practical simulations, we will explore the dynamic interactions within the power grid and the crucial role of frequency control in ensuring a resilient and reliable power system.

Parameters:

Bus:

UNTITLED														
	ID	Extra ID	Zone	Status	Volt Sol	Angle Sol	kV Base	kV Oper	pu Min	pu Max	RQ01	XQ01	RQ00	XQ00
1	B1		3	☒	0.9904103	-2.5290679	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
2	B10		1	☒	0.9504266	2.7635333	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
3	B11		1	☒	0.9454574	1.2644623	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
4	B12		3	☒	0.9239152	1.8363387	138.000	138.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
5	B13		2	☒	0.9440559	2.5302600	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
6	B14		2	☒	0.9328422	2.1199376	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
7	B15		1	☒	0.9207584	6.4032087	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
8	B16		1	☒	0.9354900	10.182559	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
9	B17		0	☒	0.9366080	7.7497396	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
10	B18		2	☒	0.9366957	5.3762912	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
11	B19		0	☒	0.9623107	22.600063	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
12	B2		1	☒	0.9675018	5.8579249	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
13	B20		4	☒	0.9591014	25.854198	230.000	230.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
14	B21		4	☒	0.9404297	13.065327	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
15	B22		4	☒	0.9668473	18.335081	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
16	B23		4	☒	0.9627920	18.092329	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
17	B24		4	☒	0.9430427	10.322582	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
18	B25		4	☒	0.9856106	8.5589380	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
19	B26		4	☒	0.9501021	10.925904	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
20	B27		4	☒	0.9366184	8.0549430	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
21	B28		4	☒	0.9369522	18.880741	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
22	B29		4	☒	0.9390324	23.483743	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
23	B3		2	☒	0.9431443	2.9070343	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
24	B30		4	☒	1.0000000	8.5386238	16.500	16.500	0.900	1.100	0.00000	0.00000	0.00000	0.00000
25	B31		4	☒	1.0000000	0.0000000	16.500	16.500	0.900	1.100	0.00000	0.00000	0.00000	0.00000
26	B32		4	☒	1.0000000	10.625136	16.500	16.500	0.900	1.100	0.00000	0.00000	0.00000	0.00000
27	B33		4	☒	1.0000000	27.837911	16.500	16.500	0.900	1.100	0.00000	0.00000	0.00000	0.00000
28	B34		4	☒	1.0000000	36.694309	16.500	16.500	0.900	1.100	0.00000	0.00000	0.00000	0.00000
29	B35		4	☒	1.0000000	23.851858	16.500	16.500	0.900	1.100	0.00000	0.00000	0.00000	0.00000
30	B36		4	☒	1.0000000	27.141803	16.500	16.500	0.900	1.100	0.00000	0.00000	0.00000	0.00000
31	B37		4	☒	1.0000000	15.834911	16.500	16.500	0.900	1.100	0.00000	0.00000	0.00000	0.00000
32	B38		4	☒	1.0000000	31.307008	16.500	16.500	0.900	1.100	0.00000	0.00000	0.00000	0.00000
33	B39		4	☒	1.0000000	-7.3410172	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
34	B4		0	☒	0.9256125	-0.9378721	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
35	B5		0	☒	0.9374627	-2.1379349	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
36	B6		0	☒	0.9414319	-1.7381973	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
37	B7		3	☒	0.9308861	-4.3188971	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
38	B8		3	☒	0.9304245	-4.9286073	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000
39	B9		0	☒	0.9827126	-6.4567794	345.000	345.000	0.900	1.100	0.00000	0.00000	0.00000	0.00000

Bus
 Fixed-Tap Xmer
 Line
 Static Load
 Generator

Transformer:

UNTITLED

	ID	Bus From	Bus To	DBase ID	Duplic	Status	Tap	Set Tap on Secondary	Min Tap	Max Tap	Number of Tap	Apply Delta Grounding	R [ohm]
1	T 01	B12	B11	T 0102	1	☒	100.00	☐	90	110	99	☐	0
2	T 02	B12	B13	T 0102	1	☒	100.00	☐	90	110	99	☐	0
3	T 03	B6	B31	T 03	1	☒	100.00	☐	90	110	99	☐	0
4	T 04	B10	B32	T 04	1	☒	100.00	☐	90	110	99	☐	0
5	T 05	B19	B33	T 05	1	☒	100.00	☐	90	110	99	☐	0
6	T 06	B20	B34	T 06	2	☒	100.00	☐	90	110	99	☐	0
7	T 07	B22	B35	T 07	1	☒	100.00	☐	90	110	99	☐	0
8	T 08	B23	B36	T 08	1	☒	100.00	☐	90	110	99	☐	0
9	T 09	B25	B37	T 09	1	☒	100.00	☐	90	110	99	☐	0
10	T 10	B2	B30	T 10	1	☒	100.00	☐	90	110	99	☐	0
11	T 11	B29	B38	T 11	1	☒	100.00	☐	90	110	99	☐	0
12	T 12	B19	B20	T 12	1	☒	100.00	☐	90	110	99	☐	0
13													
14													
15													
16													
17													
18													
19													
20													
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23													
24													
25													
26													
27													
28													
29													
30													

Bus

Fixed-Tap Xmer

Line

Static Load

Generator

Line:

UNTITLED								
	ID	Bus From	Bus To	DBase ID	Duplic	Status	Series X Compensati on	Length
1	L1	B1	B2	B12_	1	✓	0.00	275.5000
2	L10	B5	B8	B58_	1	✓	0.00	75.1000
3	L11	B6	B7	B67_	1	✓	0.00	61.7000
4	L12	B6	B11	B611_	1	✓	0.00	55.0000
5	L13	B7	B8	B78_	1	✓	0.00	30.8000
6	L14	B8	B9	B89_	1	✓	0.00	243.3000
7	L15	B9	B39	B939_	1	✓	0.00	167.6000
8	L16	B10	B11	B1011_	1	✓	0.00	28.8000
9	L17	B10	B13	B1013_	1	✓	0.00	28.8000
10	L18	B13	B14	B1314_	1	✓	0.00	67.7000
11	L19	B14	B15	B1415_	1	✓	0.00	145.4000
12	L2	B1	B39	B139_	1	✓	0.00	167.6000
13	L20	B15	B16	B1516_	1	✓	0.00	63.0000
14	L21	B16	B17	B1617_	1	✓	0.00	59.7000
15	L22	B16	B19	B1619_	1	✓	0.00	130.7000
16	L23	B16	B21	B1621_	1	✓	0.00	90.5000
17	L24	B16	B24	B1624_	1	✓	0.00	39.5000
18	L25	B17	B18	B1718_	1	✓	0.00	55.0000
19	L26	B17	B27	B1727_	1	✓	0.00	116.0000
20	L27	B21	B22	B2122_	1	✓	0.00	93.8000
21	L28	B22	B23	B2223_	1	✓	0.00	64.3000
22	L29	B23	B24	B2324_	1	✓	0.00	234.6000
23	L3	B2	B3	B23_	1	✓	0.00	101.2000
24	L30	B25	B26	B2526_	1	✓	0.00	216.5000
25	L31	B26	B27	B2627_	1	✓	0.00	98.5000
26	L32	B26	B28	B2628_	1	✓	0.00	317.7000
27	L33	B26	B29	B2629_	1	✓	0.00	418.9000
28	L34	B28	B29	B2829_	1	✓	0.00	101.2000
29	L4	B2	B25	B225_	1	✓	0.00	57.6000
30	L5	B3	B4	B34_	1	✓	0.00	142.8000
31	L6	B3	B18	B318_	1	✓	0.00	89.1000
32	L7	B4	B5	B45_	1	✓	0.00	85.8000
33	L8	B4	B14	B414_	1	✓	0.00	86.5000
34	L9	B5	B6	B56_	1	✓	0.00	17.4000
35								
36								
37								
Bus		Fixed-Tap Xmer		Line		Static Load		Generator

Static Load:

UNTITLED															
	ID	Bus ID	DBase ID	Load %	Status	P Load	Q Load	MVA	P. Factor	Extra ID 1	Extra ID 2	Extra ID 3	Creation Date	nP	nQ
1	L 01	B3	L ID 1	100	✓	322.0000	2.4000	322.0089	1.0000				4 /28/2024	0.000	0.000
2	L 02	B4	L ID 2	100	✓	500.0000	184.0000	532.7814	0.9385				4 /28/2024	0.000	0.000
3	L 03	B7	L ID 3	100	✓	233.8000	84.0000	248.4320	0.9411				4 /28/2024	0.000	0.000
4	L 04	B8	L ID 4	100	✓	522.0000	176.0000	550.8720	0.9476				4 /28/2024	0.000	0.000
5	L 05	B12	L ID 5	100	✓	7.5000	88.0000	88.3190	0.0849				4 /28/2024	0.000	0.000
6	L 06	B15	L ID 6	100	✓	320.0000	153.0000	354.6956	0.9022				4 /28/2024	0.000	0.000
7	L 07	B16	L ID 7	100	✓	329.0000	32.3000	330.5818	0.9952				4 /28/2024	0.000	0.000
8	L 08	B18	L ID 8	100	✓	158.0000	30.0000	160.8229	0.9824				4 /28/2024	0.000	0.000
9	L 09	B20	L ID 9	100	✓	628.0000	103.0000	636.3906	0.9868				4 /28/2024	0.000	0.000
10	L 10	B21	L ID 10	100	✓	274.0000	115.0000	297.1548	0.9221				4 /28/2024	0.000	0.000
11	L 11	B23	L ID 11	100	✓	247.5000	84.6000	261.5596	0.9462				4 /28/2024	0.000	0.000
12	L 12	B24	L ID 12	100	✓	308.6000	-92.2000	322.0789	0.9582				4 /28/2024	0.000	0.000
13	L 13	B25	L ID 13	100	✓	224.0000	47.2000	228.9189	0.9785				4 /28/2024	0.000	0.000
14	L 14	B26	L ID 14	100	✓	139.0000	17.0000	140.0357	0.9926				4 /28/2024	0.000	0.000
15	L 15	B27	L ID 15	100	✓	281.0000	75.5000	290.9661	0.9657				4 /28/2024	0.000	0.000
16	L 16	B28	L ID 16	100	✓	206.0000	27.6000	207.8407	0.9911				4 /28/2024	0.000	0.000
17	L 17	B39	L ID 17	100	✓	283.5000	26.9000	284.7733	0.9955				4 /28/2024	0.000	0.000
18	L 18	B31	L ID 18	100	✓	9.2000	4.6000	10.2859	0.8944				4 /28/2024	0.000	0.000
19	L 19	B39	L ID 19	100	✓	1104.0000	250.0000	1131.9523	0.9753				4 /28/2024	0.000	0.000
20															
21															
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30															

Bus
 Fixed-Tap Xmer
 Line
 Static Load
 Generator

Generator:

UNTITLED													
	ID	Bus ID	DBase ID	Duplic	Status	Generator Type	P Gen	Q Desired	Rotor Angle	Ground R	Ground X	Ctrl'd BusID	Rating
1	G01	B39	G0 ID 1	1	✓	Voltage	1000.000	0.000	0.00	9999.000	9999.000	B39	10000
2	G010	B30	G0 ID 10	1	✓	Voltage	250.000	0.000	0.00	9999.000	9999.000	B30	1000
3	G02	B31	G0 ID 2	1	✓	Swing	0.000	0.000	0.00	9999.000	9999.000	B31	700
4	G03	B32	G0 ID 3	1	✓	Voltage	650.000	0.000	0.00	9999.000	9999.000	B32	800
5	G04	B33	G0 ID 4	1	✓	Voltage	632.000	0.000	0.00	9999.000	9999.000	B33	800
6	G05	B34	G0 ID 5	2	✓	Voltage	508.000	0.000	0.00	9999.000	9999.000	B34	300
7	G06	B35	G0 ID 6	1	✓	Voltage	650.000	0.000	0.00	9999.000	9999.000	B35	800
8	G07	B36	G0 ID 7	1	✓	Voltage	560.000	0.000	0.00	9999.000	9999.000	B36	700
9	G08	B37	G0 ID 8	1	✓	Voltage	540.000	0.000	0.00	9999.000	9999.000	B37	700
10	G09	B38	G0 ID 9	1	✓	Voltage	830.000	0.000	0.00	9999.000	9999.000	B38	1000
11													
12													
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29													
30													

Bus
 Fixed-Tap Xmer
 Line
 Static Load
 Generator

Power Flow Analysis:

Load Flow Solver Dialog

Method	Options	Maximum # iterations
☐ Fast Decoupled	☒ Flat start	60
☐ Gauss Seidel	☐ Apply constraints	
☒ Newton-Raphson	☒ Fix tap for Tcul txfo	

☒ Block Q-flow Txfo Adjustment	☒ Block P-flow Txfo Adjustment
☒ Block Switchable Shunt Adjustment	☒ Block DC Link Adjustment

☐ Area Interchange	☐ Distributed Swing
---	--

☒ Keep iterations	☒ Create input file for CymStab
---	---

Solve!

OK

Cancel

Study...

Units...

Iteration Report:

UNTITLED: Iteration Report													
MISMATCHES						ADJUSTMENTS							
Iter#	dP	Bus	dQ	Bus	Area	Gen	TCV	TCQ	TCP	DCL	SvS		
1	.1016E+02	B34		.1593E+01	B4		0/0	0/0	0/0	0/0	0/0	0/0	0/0
2	.4384E+00	B8		.1378E+01	B19		0/0	0/0	0/0	0/0	0/0	0/0	0/0
3	.3730E-01	B6		.5846E-01	B16		0/0	0/0	0/0	0/0	0/0	0/0	0/0
4	.2245E-03	B6		.1884E-03	B16		0/0	0/0	0/0	0/0	0/0	0/0	0/0
*** NEWTON-RAPHSON I.F. CONVERGED ***													
save is ok													
save is ok													

After performing Power Flow Analysis, we got the result to be converged with Newton Raphson numerical method.

Now we move to performing the Frequency Response Analysis for the outage of generator G08 (i.e. loss of 540 MW generation).

Frequency Response Analysis:

Kp = 0%

Transient Stability Study Dialog

Events	Monitoring	Real time plotting	Tabular report	Relay/Zones/Areas	Display Options		
Title	Settings	LF data	Global Load	Load/Zone	Frequency/Zone	Global Selection	UDM

Zload Voltage below which load is represented by const Z

nP Exponent of voltage term in active load function

nQ Exponent of voltage term in reactive load function

Pfreq Correction factor in function: Load MW vs Frequency

Qfreq Correction factor in function: Load MVAR vs Frequency

Default

Reload

OK CANCEL APPLY

Generator removal: Generator 08 (540 MW)

Transient Stability Study Dialog

Title | Settings | LF data | Global Load | Load/Zone | Frequency/Zone | Global Selection | UDM
 Events | Monitoring | Real time plotting | Tabular report | Relay/Zones/Areas | Display Options

Events view

10.0 Generator Removal	G08
------------------------	-----

Edit

OK CANCEL APPLY

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM
 Database ID: DEF_VALUES Access DB
 Connected to bus: GlobalRelay Make Available To

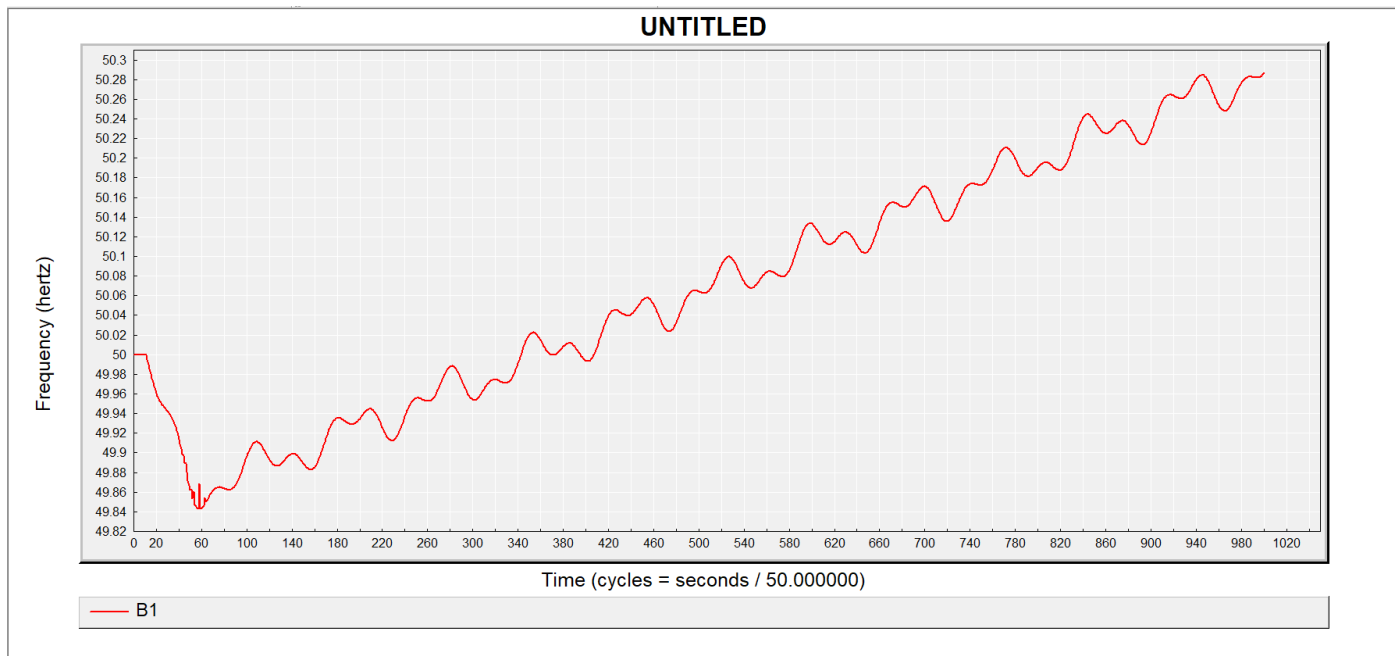
New
 Delete
 Reload
 Default
 Help

First frequency threshold [Hz]:	49.90000
First load shedding amount [%]:	10.00000
Second frequency threshold [Hz]:	49.00000
Second load shedding amount [%]:	25.00000
Third frequency threshold [Hz]:	48.00000
Third load shedding amount [%]:	35.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

OK CANCEL << >>

Simulation:



We found the Nadir Point to be 49.84

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

10.00000

Second frequency threshold [Hz]:

49.30000

Second load shedding amount [%]:

20.00000

Third frequency threshold [Hz]:

48.00000

Third load shedding amount [%]:

40.00000

Relay operating time [sec.]:

0.10000

EQ / TOTAL

1 / 1

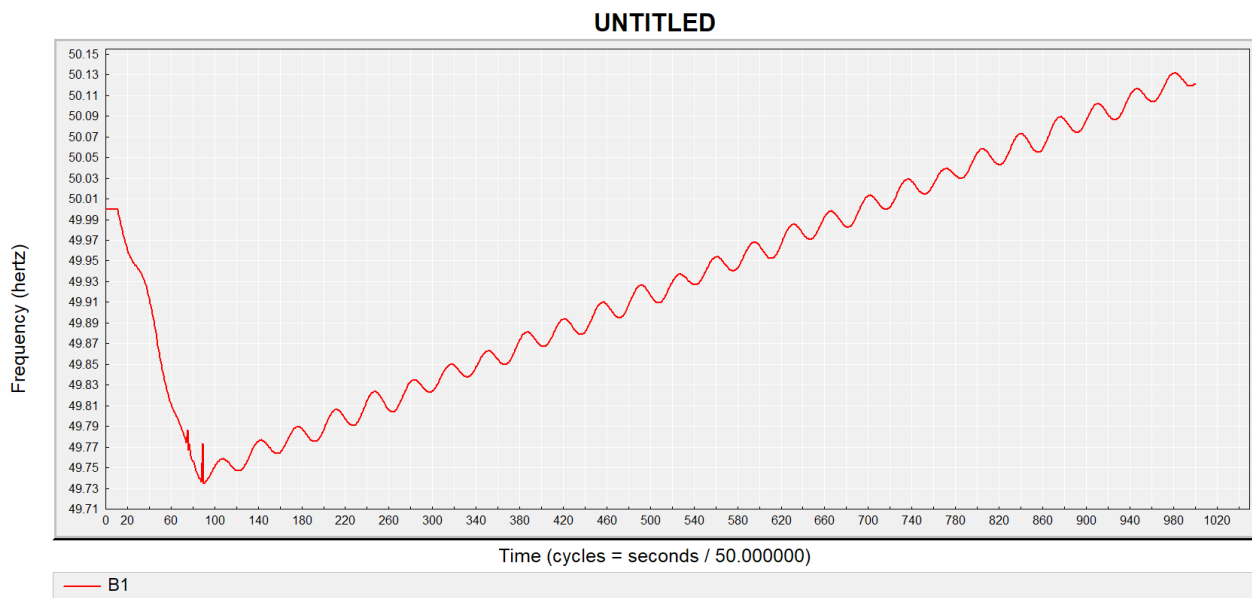
OK

CANCEL

<<

>>

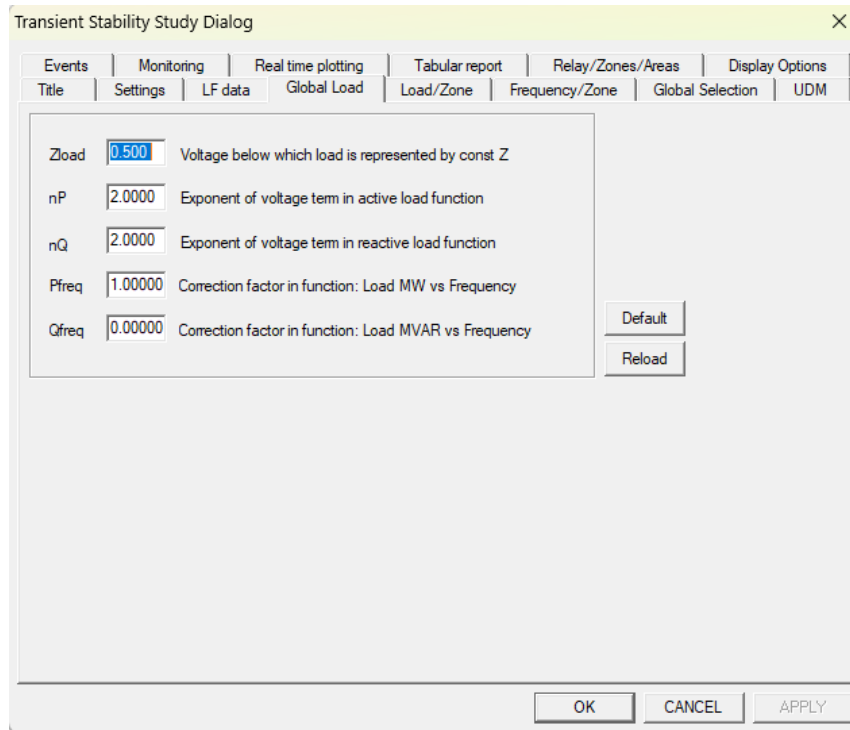
Simulation:



We found the Nadir Point to be 49.734

Now we re-do the simulation for different K_p values.

$K_p = 1\%$



The image shows a software dialog box titled "Transient Stability Study Dialog". It features a tabbed interface with the following tabs: Events, Monitoring, Real time plotting, Tabular report, Relay/Zones/Areas, Display Options, Title, Settings, LF data, Global Load, Load/Zone, Frequency/Zone, Global Selection, and UDM. The "Settings" tab is currently selected. Inside the dialog, there are five input fields with their corresponding descriptions:

- Zload**: 0.5000. Description: Voltage below which load is represented by const Z.
- nP**: 2.0000. Description: Exponent of voltage term in active load function.
- nQ**: 2.0000. Description: Exponent of voltage term in reactive load function.
- Pfreq**: 1.00000. Description: Correction factor in function: Load MW vs Frequency.
- Qfreq**: 0.00000. Description: Correction factor in function: Load MVAR vs Frequency.

There are two buttons on the right side of the input fields: "Default" and "Reload". At the bottom of the dialog, there are three buttons: "OK", "CANCEL", and "APPLY".

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay

UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

First frequency threshold [Hz]:

49.90000

First load shedding amount [%]:

10.00000

Second frequency threshold [Hz]:

49.00000

Second load shedding amount [%]:

25.00000

Third frequency threshold [Hz]:

48.00000

Third load shedding amount [%]:

35.00000

Relay operating time [sec.]:

0.10000

New

Delete

Reload

Default

Help

EQ / TOTAL

1 / 1

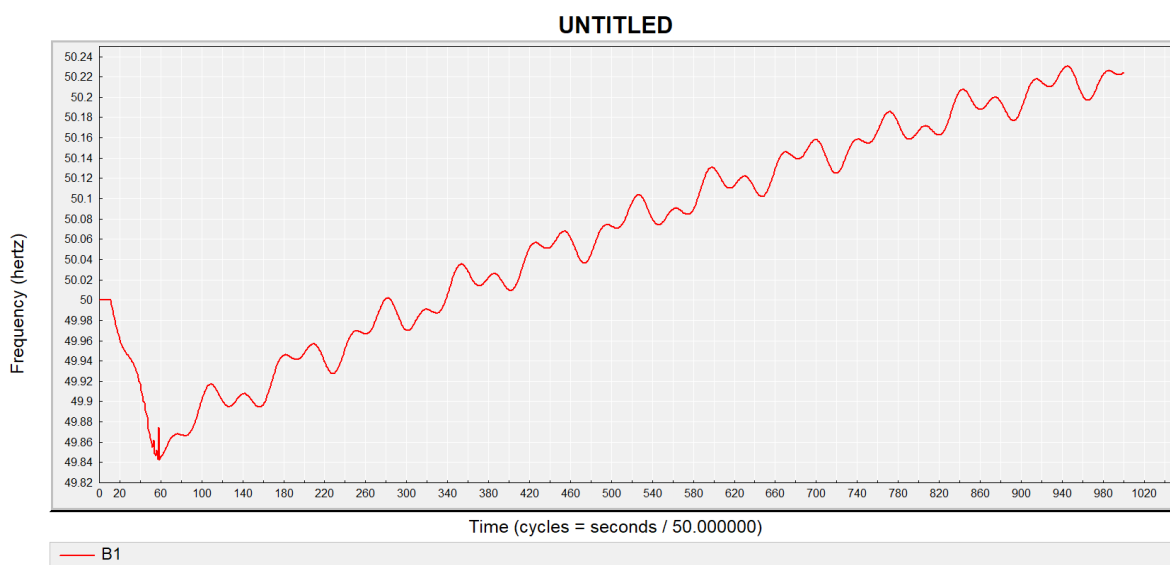
OK

CANCEL

<<

>>

Simulation:



We found the Nadir Point to be 49.84

Kp = 2%:

Transient Stability Study Dialog

Events	Monitoring	Real time plotting	Tabular report	Relay/Zones/Areas	Display Options
Title	Settings	LF data	Global Load	Load/Zone	Frequency/Zone
				Global Selection	UDM

Zload	0.500	Voltage below which load is represented by const Z
nP	2.0000	Exponent of voltage term in active load function
nQ	2.0000	Exponent of voltage term in reactive load function
Pfreq	2.00000	Correction factor in function: Load MW vs Frequency
Qfreq	0.00000	Correction factor in function: Load MVAR vs Frequency

Default
Reload

OK CANCEL APPLY

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay UDM

Low Freq. ID	NEW ITEM	
Database ID	DEF_VALUES	Access DB
Connected to bus:	Global/Relay	Make Available To

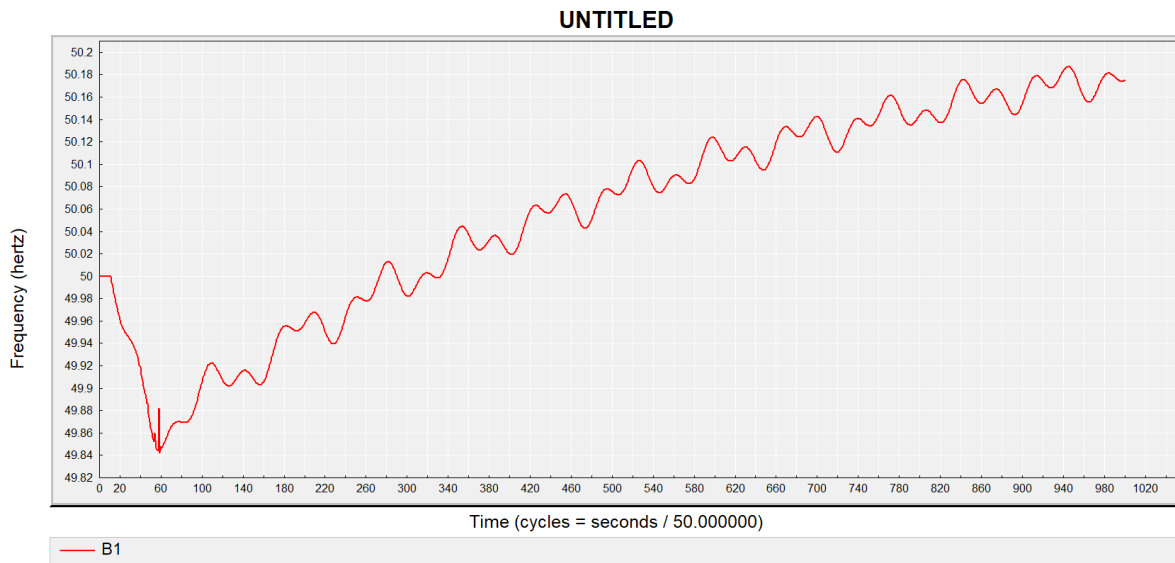
First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	10.00000
Second frequency threshold [Hz]:	49.30000
Second load shedding amount [%]:	20.00000
Third frequency threshold [Hz]:	48.00000
Third load shedding amount [%]:	40.00000
Relay operating time [sec.]:	0.10000

New
Delete
Reload
Default
Help

EQ / TOTAL
1 / 1

OK CANCEL << >>

Simulation:



We found the Nadir Point to be 49.84

Kp = 3%:

Transient Stability Study Dialog

Events	Monitoring	Real time plotting	Tabular report	Relay/Zones/Areas	Display Options		
Title	Settings	LF data	Global Load	Load/Zone	Frequency/Zone	Global Selection	UDM
Zload	0.500	Voltage below which load is represented by const Z					
nP	2.0000	Exponent of voltage term in active load function					
nQ	2.0000	Exponent of voltage term in reactive load function					
Pfreq	3.00000	Correction factor in function: Load MW vs Frequency					
Qfreq	0.00000	Correction factor in function: Load MVAR vs Frequency					
						Default	
						Reload	

OK CANCEL APPLY

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay

UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

10.00000

Second frequency threshold [Hz]:

49.30000

Second load shedding amount [%]:

20.00000

Third frequency threshold [Hz]:

48.00000

Third load shedding amount [%]:

40.00000

Relay operating time [sec.]:

0.10000

EQ / TOTAL

1 / 1

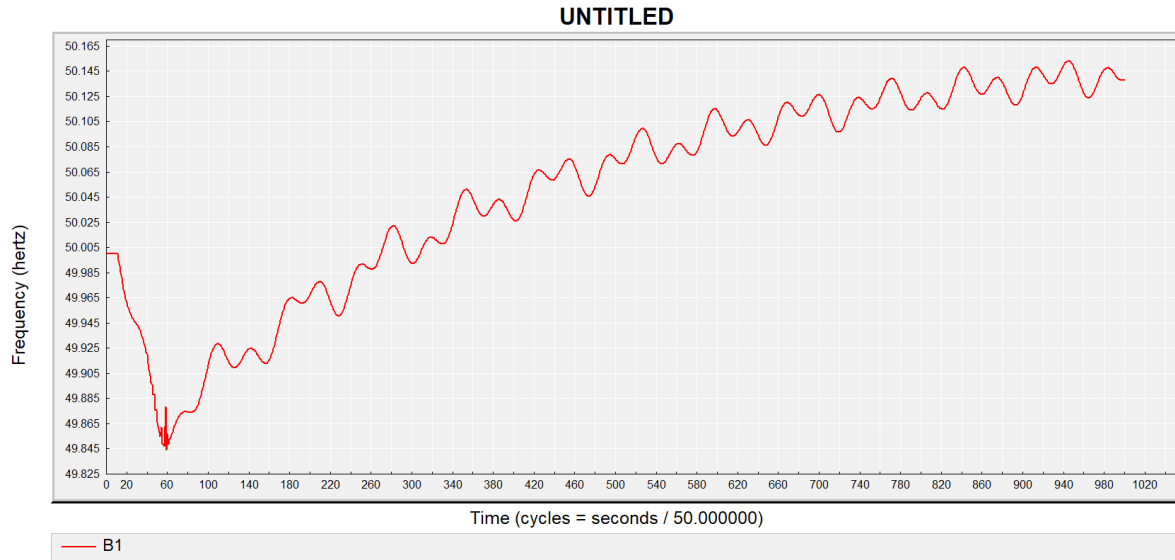
OK

CANCEL

<<

>>

Simulation:



We found the Nadir Point to be 49.845.

FVSI Calculation:

The Fast Voltage Stability Index (FVSI) is a critical parameter used in power systems to assess the stability of voltage levels within the network. FVSI is a numerical indicator that helps in identifying potential voltage instability or collapse points in a power system. It is calculated based on system parameters such as line reactance, load power, and sending-end voltage. The significance of FVSI lies in its ability to provide a quick and reliable assessment of voltage stability, enabling operators to take preemptive measures to prevent voltage collapse. By monitoring FVSI values, power system operators can identify weak points in the grid and implement corrective actions such as adjusting reactive power support or reconfiguring the network. This proactive approach helps in maintaining system reliability, preventing blackouts, and ensuring the continuous delivery of electric power to consumers.

Formula:

$$FVSI_{zk} = \frac{4Z^2 Q_k}{V_S^2 X}$$

Code:

We generate a Matlab code for determining the FVSI for each load bus.

```
clear all
close all
vs=345;
i=0;
i=i+1;
z(i)=0.0013+j*0.151 ;
q(i)=2.4 ;
x(i)=0.151 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0013+j*0.0213 ;
q(i)=184 ;
x(i)=0.0213 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0006+j*0.0092 ;
q(i)=84 ;
x(i)=0.0092 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0008+j*0.0112 ;
q(i)=176 ;
x(i)=0.0112 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0048+j*0.1305 ;
q(i)=88 ;
x(i)=0.1305 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0018+j*0.0217 ;
q(i)=153 ;
x(i)=0.0217 ;
```

```

fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0009+j*0.0094 ;
q(i)=32.3 ;
x(i)=0.0094 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0007+j*0.0082 ;
q(i)=30 ;
x(i)=0.0082 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.007+j*0.138 ;
q(i)=103 ;
x(i)=0.138 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0008+j*0.0135 ;
q(i)=115 ;
x(i)=0.0135 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0006+j*0.0096 ;
q(i)=84.6 ;
x(i)=0.0096 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0022+j*0.035 ;
q(i)=-92.2 ;
x(i)=0.035 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.007+j*0.0086 ;
q(i)=47.2 ;
x(i)=0.0086 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0032+j*0.0323 ;
q(i)=17 ;
x(i)=0.0323 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0014+j*0.0147 ;
q(i)=75.5 ;
x(i)=0.0147 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));

```

```

i=i+1;
z(i)=0.0043+j*0.0474 ;
q(i)=27.6 ;
x(i)=0.0474 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.0057+j*0.0625 ;
q(i)=26.9 ;
x(i)=0.0625 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.00001+j*0.175 ;
q(i)=4.6 ;
x(i)=0.175 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));
i=i+1;
z(i)=0.001+j*0.025 ;
q(i)=250 ;
x(i)=0.025 ;
fvsi(i)=(4*(abs(z(i))^2)*q(i))/((vs^2)*x(i));

sort_fvsi=sort(fvsi);
for k=1:19

    a=(sort_fvsi(k)==fvsi);
    sort_load(k)=find(a);

end

sort_load=sort_load'
sort_fsvi=sort_fvsi'

t=table(sort_load,sort_fsvi)

```

Sorted Output of the Code fro, stronger bus to weaker bus:

	sort_load	sort_fsvi
1	12	-1.0888e-04
2	8	8.3274e-06
3	7	1.0297e-05
4	1	1.2180e-05
5	14	1.8634e-05
6	13	2.2679e-05
7	3	2.6081e-05
8	18	2.7053e-05
9	11	2.7400e-05
10	15	3.7636e-05
11	16	4.4327e-05
12	10	5.2357e-05
13	17	5.6971e-05
14	4	6.6583e-05
15	6	1.1234e-04
16	2	1.3220e-04
17	19	2.1038e-04
18	5	3.8646e-04
19	9	4.7891e-04

Zone Sectorization:

For better utilization of load shedding scheme, we divide the system into separate zones according to the FVSI values, where higher FVSI value for a bus signifies weaker bus and lower FVSI value signifies stronger bus.

As per the sorted table of FVSI values, we design the system such as shown below.

Zone	Loads	Buses	Attribute
0	9	20	Weakest Zone
	5	12	
	19	39	

	2	04	
	6	15	
	4	08	
	17	29	
1	10 16 15 11	21 28 27 23	Weak Zone
2	18 3 13 14	31 07 25 26	Strong Zone
3	1 7 8 12	03 16 18 24	Strongest Zone
4	-	All other Buses	Neutral

Now we design a load shedding scheme to apply more load cut to weaker buses following the loss of G08.

For $K_p = 0\%$

Transient Stability Study Dialog

Events | Monitoring | Real time plotting | Tabular report | Relay/Zones/Areas | Display Options

Title | Settings | LF data | Global Load | Load/Zone | Frequency/Zone | Global Selection | UDM

Zload: 0.500 Voltage below which load is represented by const Z

nP: 1.0000 Exponent of voltage term in active load function

nQ: 1.0000 Exponent of voltage term in reactive load function

Pfreq: 0.00000 Correction factor in function: Load MW vs Frequency

Qfreq: 0.00000 Correction factor in function: Load MVAR vs Frequency

Default

Reload

OK CANCEL APPLY

Zone 0:

Transient Stability Study Dialog

Title Settings LF data Global Load Load/Zone Frequency/Zone Global Selection UDM
Events Monitoring Real time plotting Tabular report Relay/Zones/Areas Display Options

Low Voltage Relay
List :

Low Frequency Relay
List : ☒ NEW ITEM

Relay Properties

Assign Zones or Areas

☐ Zones:

- ☒ 0
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

Data Selection

☐ When solving, override specific component data with this one.

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

Database ID:

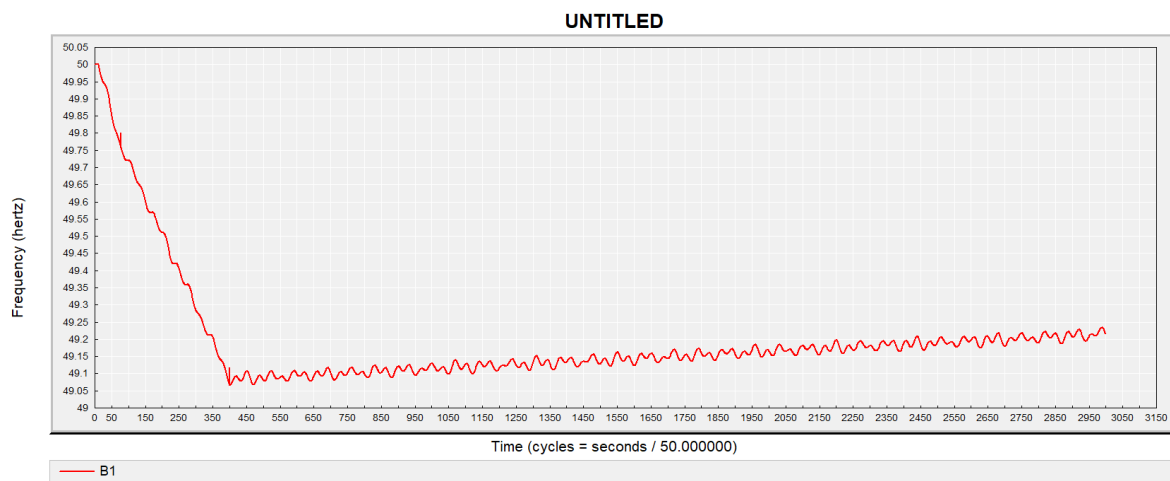
Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	60.00000
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	70.00000
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	80.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

<< >>

Simulation:



Nadir Point: 49.06 Hz

Zone 1:

Transient Stability Study Dialog

Title | Settings | LF data | Global Load | Load/Zone | Frequency/Zone | Global Selection | UDM
Events | Monitoring | Real time plotting | Tabular report | Relay/Zones/Areas | Display Options

Low Voltage Relay
List :

Low Frequency Relay
List : ☒ NEW ITEM

Relay Properties

Assign Zones or Areas

☐ Zones:

- ☐ 0
- ☒ 1
- ☐ 2
- ☐ 3
- ☐ 4

Data Selection

☐ When solving, override specific component data with this one.

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

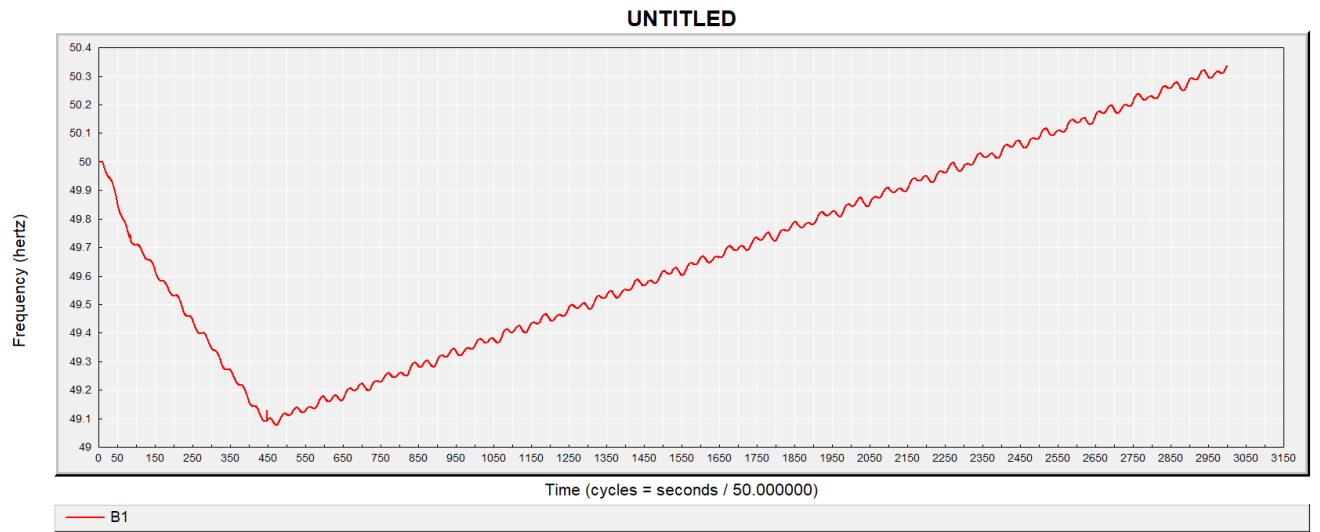
Database ID:

Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	45.00000
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	50.00000
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	60.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: 49.08Hz

Zone 2:

Transient Stability Study Dialog

Title | Settings | LF data | Global Load | Load/Zone | Frequency/Zone | Global Selection | UDM |
Events | Monitoring | Real time plotting | Tabular report | Relay/Zones/Areas | Display Options

Low Voltage Relay
List :

Low Frequency Relay
List : ☒ NEW ITEM

Relay Properties

Assign Zones or Areas

☐ Zones:

- ☐ 0
- ☐ 1
- ☒ 2
- ☐ 3
- ☐ 4

Data Selection

☐ When solving, override specific component data with this one.

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency RelayUDM

Low Freq. IDNEW ITEM

Database IDDEF_VALUESAccess DB

Connected to bus:GlobalRelayMake Available To

First frequency threshold [Hz]:49.80000

First load shedding amount [%]:43.00000

Second frequency threshold [Hz]:49.10000

Second load shedding amount [%]:47.00000

Third frequency threshold [Hz]:48.60000

Third load shedding amount [%]:58.00000

Relay operating time [sec.]:0.10000

New

Delete

Reload

Default

Help

EQ / TOTAL

1 / 1

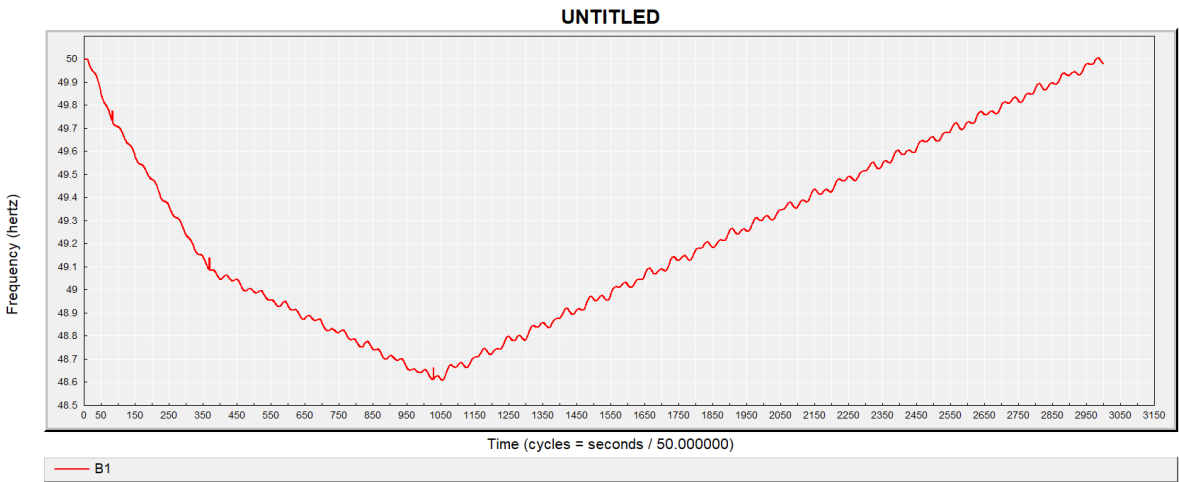
OK

CANCEL

<<

>>

Simulation:



Nadir Point: 48.6 Hz

Zone 3:

Transient Stability Study Dialog

Title | Settings | LF data | Global Load | Load/Zone | Frequency/Zone | Global Selection | UDM |
Events | Monitoring | Real time plotting | Tabular report | Relay/Zones/Areas | Display Options

Low Voltage Relay
List : Add / Edit ... Delete

Low Frequency Relay
List : ☒ NEW ITEM Add / Edit ... Delete

Relay Properties

Assign Zones or Areas

☐ Zones:

- ☐ 0
- ☐ 1
- ☐ 2
- ☒ 3
- ☐ 4

Data Selection

☐ When solving, override specific component data with this one.

OK CANCEL APPLY

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: New

Database ID: Access DB Delete

Connected to bus: Make Available To Reload

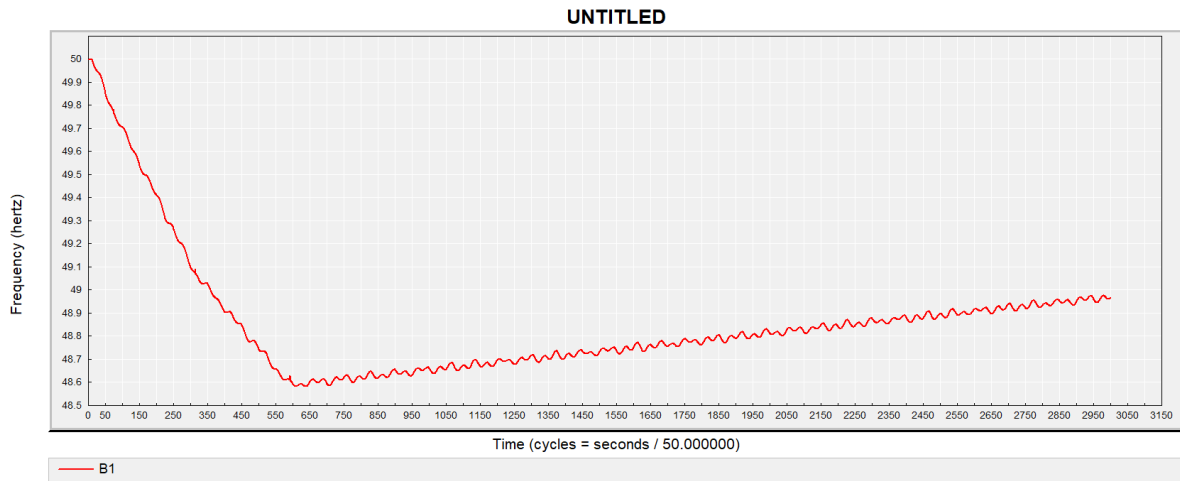
Default Help

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	20.00000
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	30.00000
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	40.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 48.57 Hz.

Analysis:

At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively. From the load shedding profile we see, for $K_p = 0\%$, as we move from the weakest zone to the strongest zone, we need to shed less load for stability of the system. So, for stronger zones, we apply less load shedding and still got an acceptable nadir frequency, where for the weaker zones we had to apply more load shedding. Though the nadir point falls a bit for the stronger zones as we apply less shedding of loads, if we increase the amount of shedding, the nadir point would increase as well.

For $K_p = 1\%$:

Transient Stability Study Dialog

Events	Monitoring	Real time plotting	Tabular report	Relay/Zones/Areas	Display Options
Title	Settings	LF data	Global Load	Load/Zone	Frequency/Zone
				Global Selection	UDM

Zload
Voltage below which load is represented by const Z

nP
Exponent of voltage term in active load function

nQ
Exponent of voltage term in reactive load function

Pfreq
Correction factor in function: Load MW vs Frequency

Qfreq
Correction factor in function: Load MVAR vs Frequency

Default
Reload

OK CANCEL APPLY

Zone 0 :

Transient Stability Study Dialog

Title	Settings	LF data	Global Load	Load/Zone	Frequency/Zone	Global Selection	UDM
Events	Monitoring	Real time plotting	Tabular report	Relay/Zones/Areas	Display Options		

Low Voltage Relay
List :
Add / Edit ...
Delete

Low Frequency Relay
List : ☒ NEW ITEM
Add / Edit ...
Delete

Relay Properties
Assign Zones or Areas

☒ Zones:

☒ 0
☐ 1
☐ 2
☐ 3
☐ 4

Data Selection
☐ When solving, override specific component data with this one.

OK CANCEL APPLY

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

First frequency threshold [Hz]: 49.80000

First load shedding amount [%]: 50.00000

Second frequency threshold [Hz]: 49.10000

Second load shedding amount [%]: 60.00000

Third frequency threshold [Hz]: 48.60000

Third load shedding amount [%]: 70.00000

Relay operating time [sec.]: 0.10000

New

Delete

Reload

Default

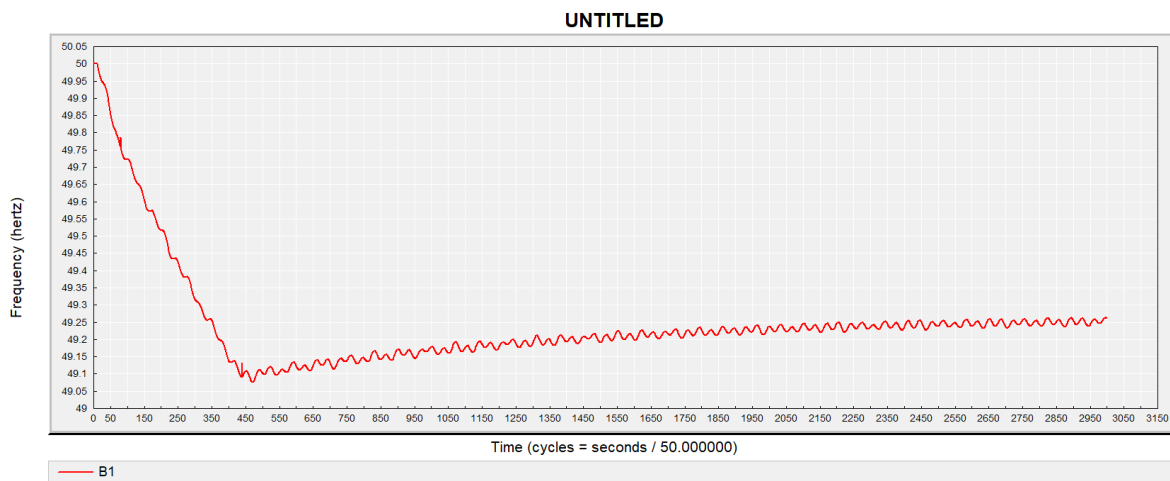
Help

EQ / TOTAL

1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 49.075 Hz.

Zone 1:

Transient Stability Study Dialog

Title | Settings | LF data | Global Load | Load/Zone | Frequency/Zone | Global Selection | UDM
Events | Monitoring | Real time plotting | Tabular report | Relay/Zones/Areas | Display Options

Low Voltage Relay
List : Add / Edit ... Delete

Low Frequency Relay
List : ☒ NEW ITEM Add / Edit ... Delete

Relay Properties

Assign Zones or Areas

☐ Zones:

- ☐ 0
- ☒ 1
- ☐ 2
- ☐ 3
- ☐ 4

Data Selection

☐ When solving, override specific component data with this one.

OK CANCEL APPLY

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

Database ID: Access DB

Connected to bus: Make Available To

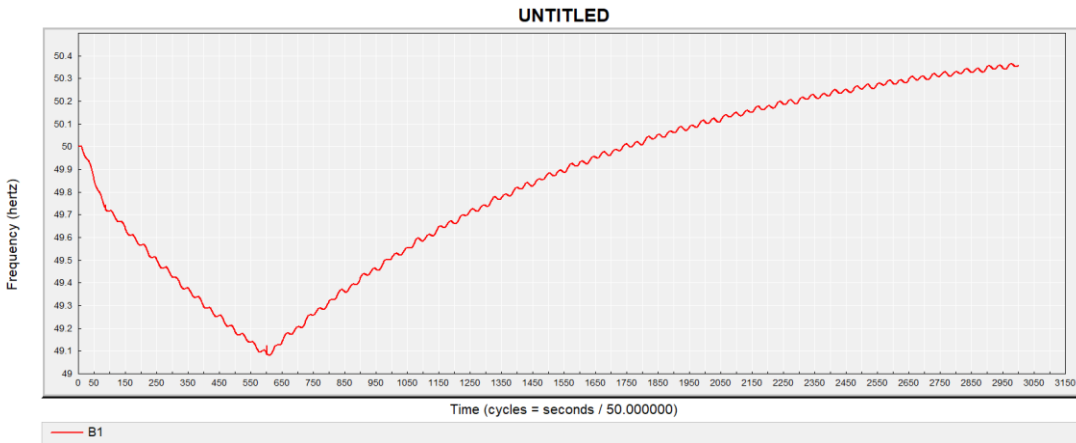
New
Delete
Reload
Default
Help

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	45.00000
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	50.00000
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	60.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 49.08 Hz.

Zone 2:

Transient Stability Study Dialog

Title | Settings | LF data | Global Load | Load/Zone | Frequency/Zone | Global Selection | UDM
Events | Monitoring | Real time plotting | Tabular report | Relay/Zones/Areas | Display Options

Low Voltage Relay

List :

Add / Edit ...
Delete

Low Frequency Relay

List : ☒ NEW ITEM

Add / Edit ...
Delete

Relay Properties

Assign Zones or Areas

☒ Zones:

☐ 0
☐ 1
☒ 2
☐ 3
☐ 4

Data Selection

☐ When solving, override specific component data with this one.

OK CANCEL APPLY

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

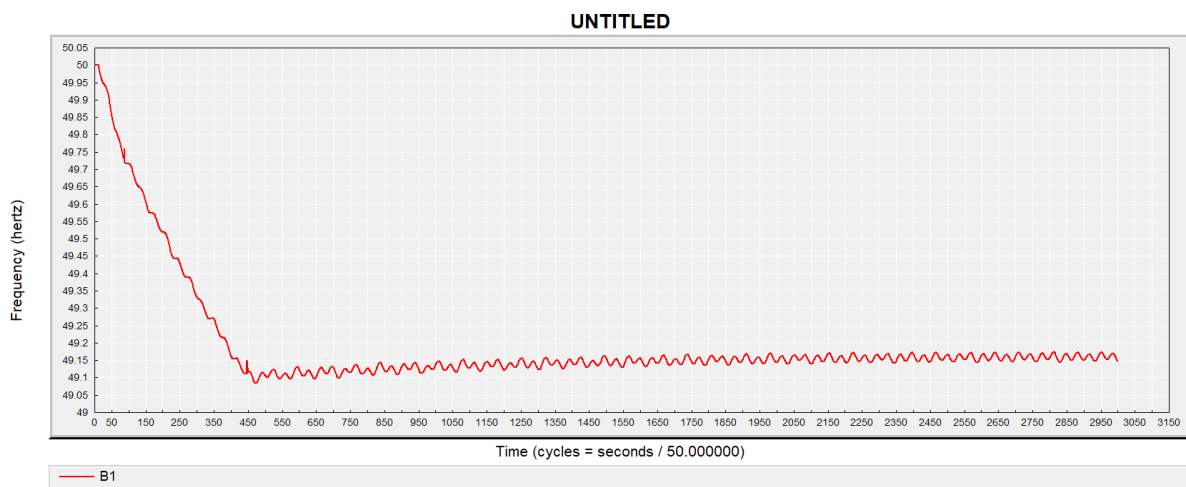
Database ID:

Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	43.00000
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	47.00000
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	58.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: 49.08 Hz.

Zone 3:

Transient Stability Study Dialog

Title Settings LF data Global Load Load/Zone Frequency/Zone Global Selection UDM
Events Monitoring Real time plotting Tabular report Relay/Zones/Areas Display Options

Low Voltage Relay
List :

Low Frequency Relay
List : ☒ NEW ITEM

Relay Properties

Assign Zones or Areas

☐ Zones:

- ☐ 0
- ☐ 1
- ☐ 2
- ☒ 3
- ☐ 4

Data Selection

☐ When solving, override specific component data with this one.

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

20.00000

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

30.00000

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

40.00000

Relay operating time [sec.]:

0.10000

EQ / TOTAL

1 / 1

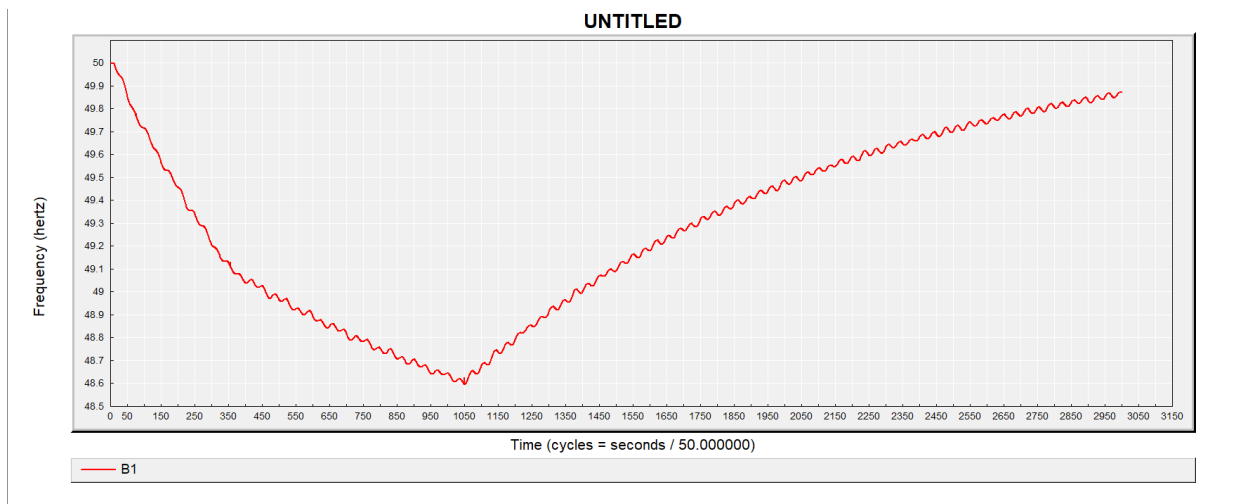
OK

CANCEL

<<

>>

Simulation:



Nadir Point: 48.59 Hz.

Analysis:

The analysis for $K_p = 1\%$ is quite similar to the $K_p = 0\%$. At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively. From the load shedding profile we see, for $K_p = 1\%$, as we move from the weakest zone to the strongest zone, we need to shed less load for stability of the system. So, for stronger zones, we apply less load shedding and still got an acceptable nadir frequency, where for the weaker zones we had to apply more load shedding. Though the nadir point falls a bit for the strongest zone (zone 3) as we apply less shedding of loads, if we increase the amount of shedding, the nadir point would increase as well.

For zone 0, the effect of changing K_p is clear, as for $K_p = 1\%$ the nadir point was 49.075 Hz even for a slight less shed amount of 50%,60% and 70% compared to $K_p = 0\%$ the nadir to be 49.06 Hz for 60%,70% and 80%.

For zone 1, the effect isn't rigid as the nadir is 49.08Hz for both $K_p = 0\%$ and 1% both same shedding profile.

But for zone 2, the increasement of K_p from 0% to 1% increases the nadir point from 48.6Hz to 49.08Hz for same shedding profile.

For zone 3, the effect of K_p again becomes vague as the nadir point is quite same for both K_p cases (48.57Hz for $K_p = 0\%$ and 48.59Hz for $K_p = 1\%$) for same shedding profile.

For $K_p = 2\%$:

Parameter	Value	Description
Zload	0.5000	Voltage below which load is represented by const Z
nP	2.0000	Exponent of voltage term in active load function
nQ	2.0000	Exponent of voltage term in reactive load function
Pfreq	2.00000	Correction factor in function: Load MW vs Frequency
Qfreq	0.00000	Correction factor in function: Load MVAR vs Frequency

Zone 0 : Case 1:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

30.00000

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

40.00000

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

50.00000

Relay operating time [sec.]:

0.10000

EQ / TOTAL

1 / 1

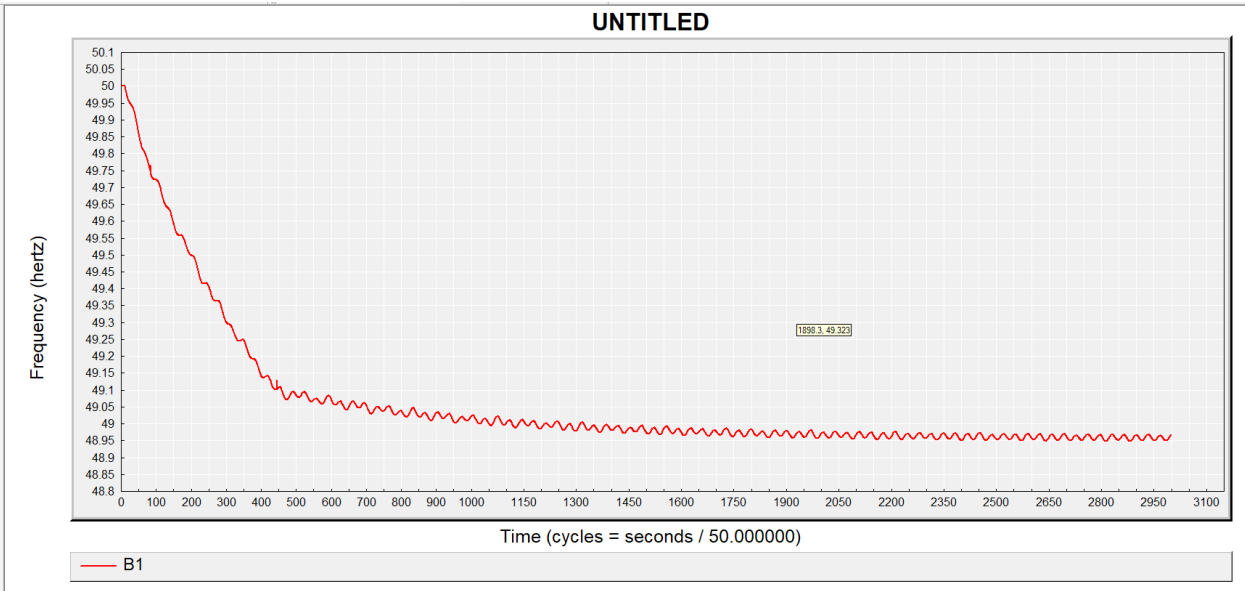
OK

CANCEL

<<

>>

Simulation:



Nadir Point: Unstable so no nadir point found.

Zone 0 : Case 2:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

50.00000

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

60.00000

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

70.00000

Relay operating time [sec.]:

0.10000

EQ / TOTAL

1 / 1

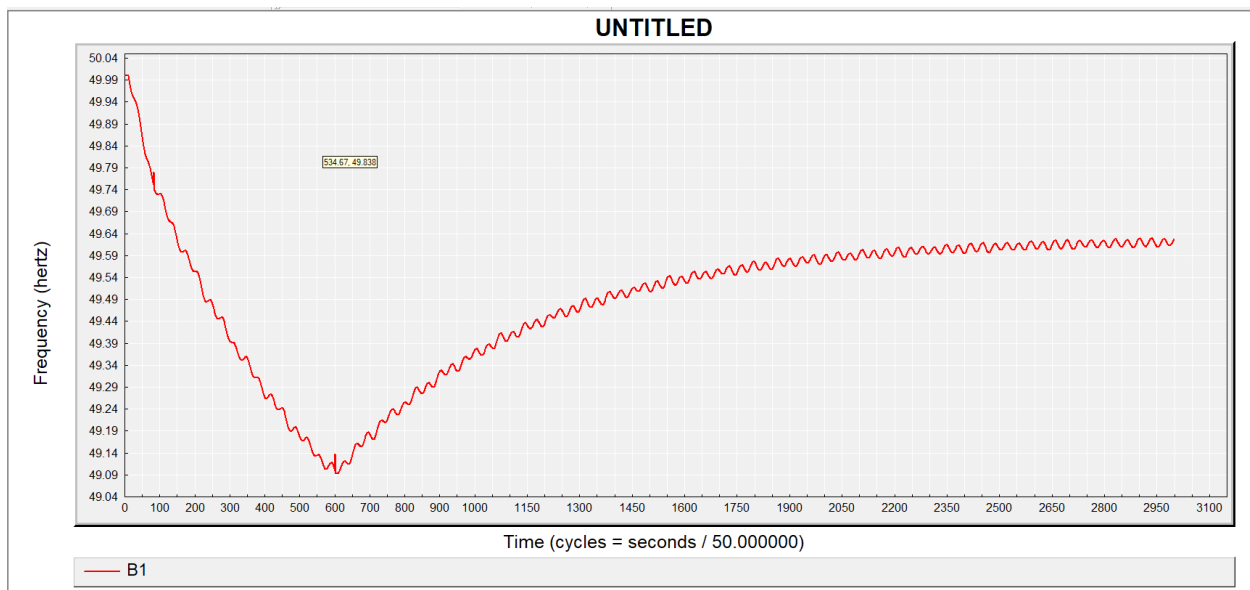
OK

CANCEL

<<

>>

Simulation:



Nadir Point: 49.09 Hz.

Zone 0 : Case 3:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay

UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

40.00000

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

50.00000

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

60.00000

Relay operating time [sec.]:

0.10000

EQ / TOTAL

1 / 1

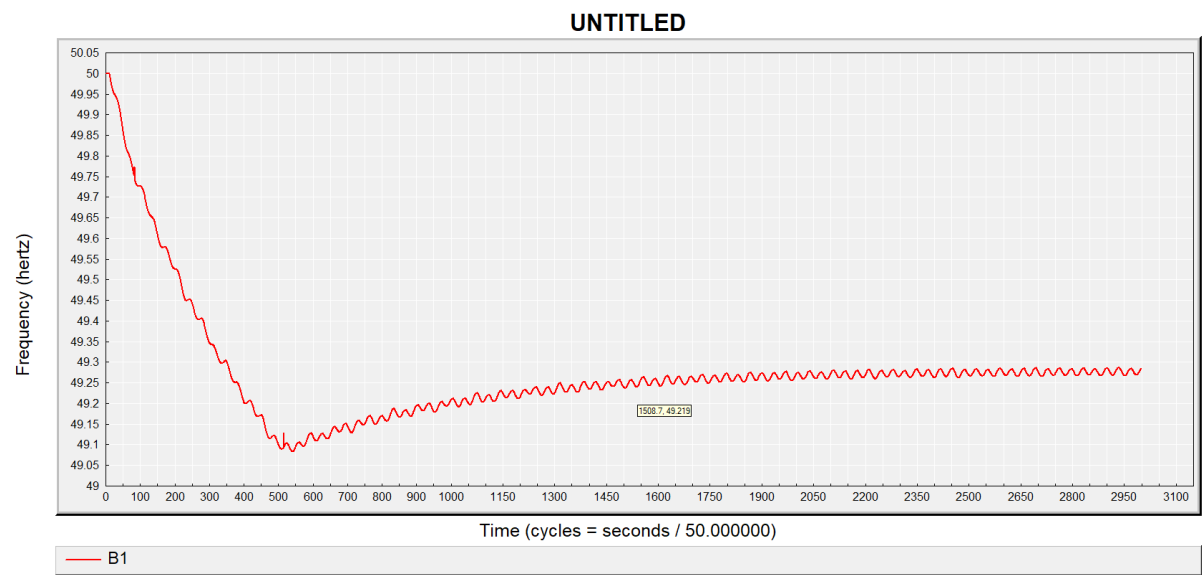
OK

CANCEL

<<

>>

Simulation:



Nadir Point: 49.075 Hz.

Analysis:

At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively. For zone 0, we analyzed the system for 3 different load shedding profiles. For the 1st case, the load shedding was less (30%,40%, and 50% for 1st,2nd and 3rd threshold) and we found from the simulation that the system is not stable and there is no nadir point.

When we increased the shed amount for the 2nd and 3rd case, we were able to make the system stable. For 2nd case, the shed amount was 50%, 60% and 70% and we got the nadir point to be 49.09 Hz. For the 3rd case, we decrease the shed amount to 40%,50% and 60% and got the nadir to be also decreased a bit to 49.075 Hz.

In comparison with the previous Kp value to be 1%, for the shedding profile to be 50%,60% and 70%, the nadir was 49.075Hz, but here in this case of Kp=2%, the nadir increased to 49.09Hz.

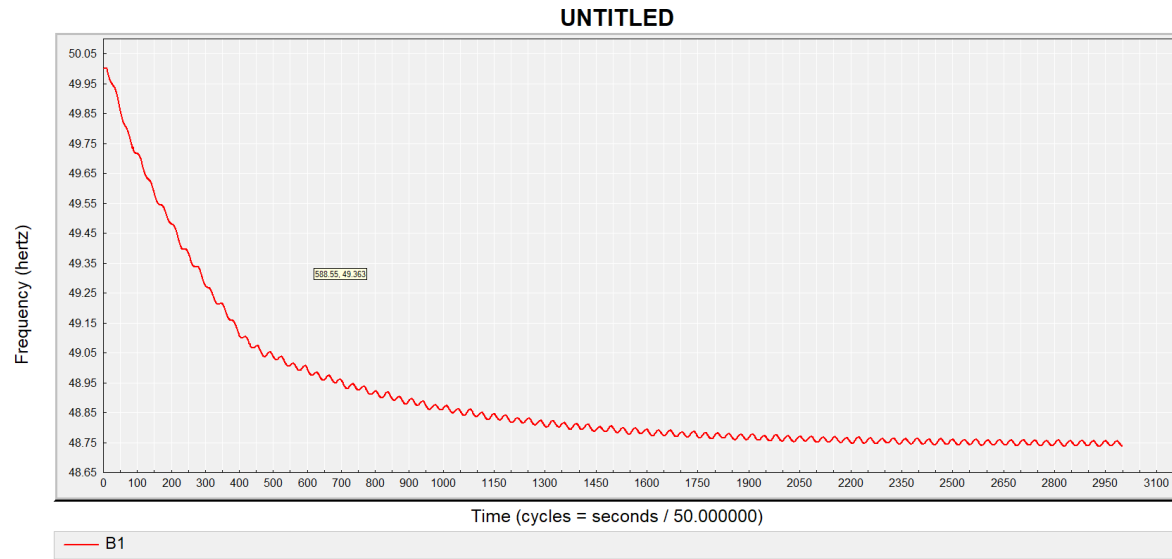
Zone 1: Case 1:

Load Shedding Profile:

Global Low Frequency Relay	
Low Frequency Relay UDM	
Low Freq. ID	NEW ITEM
Database ID	DEF_VALUES
Connected to bus:	GlobalRelay
First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	15.00000
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	20.00000
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	30.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: Unstable so no nadir point found.

Zone 1: Case 2:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

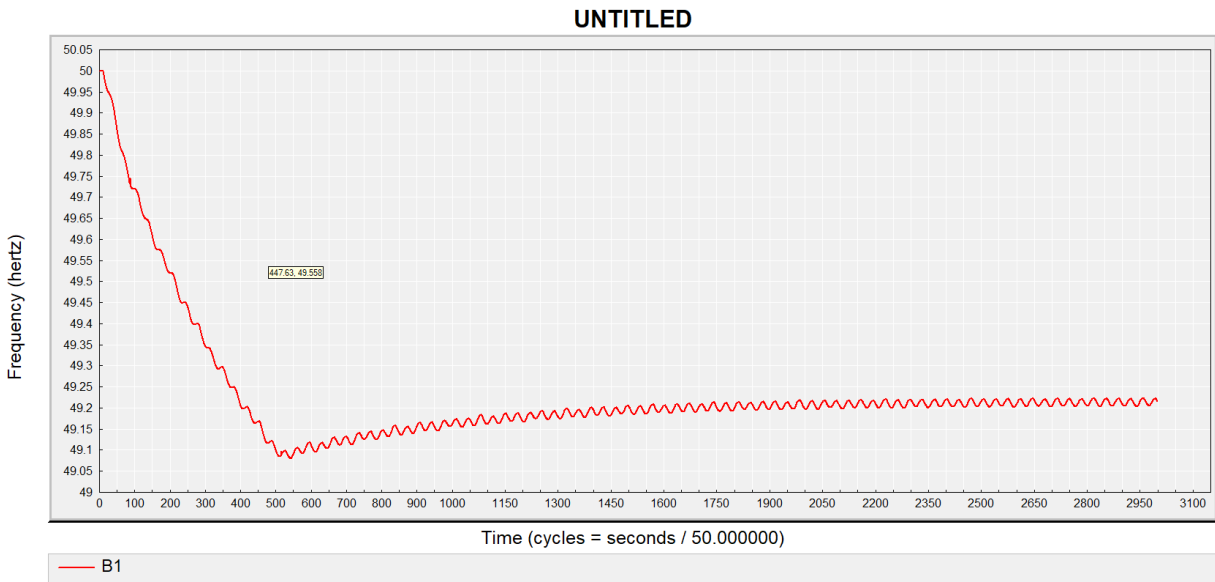
Database ID:

Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	25.00000
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	30.00000
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	40.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: 49.075 Hz.

Zone 1: Case 3:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

New

Delete

Reload

Default

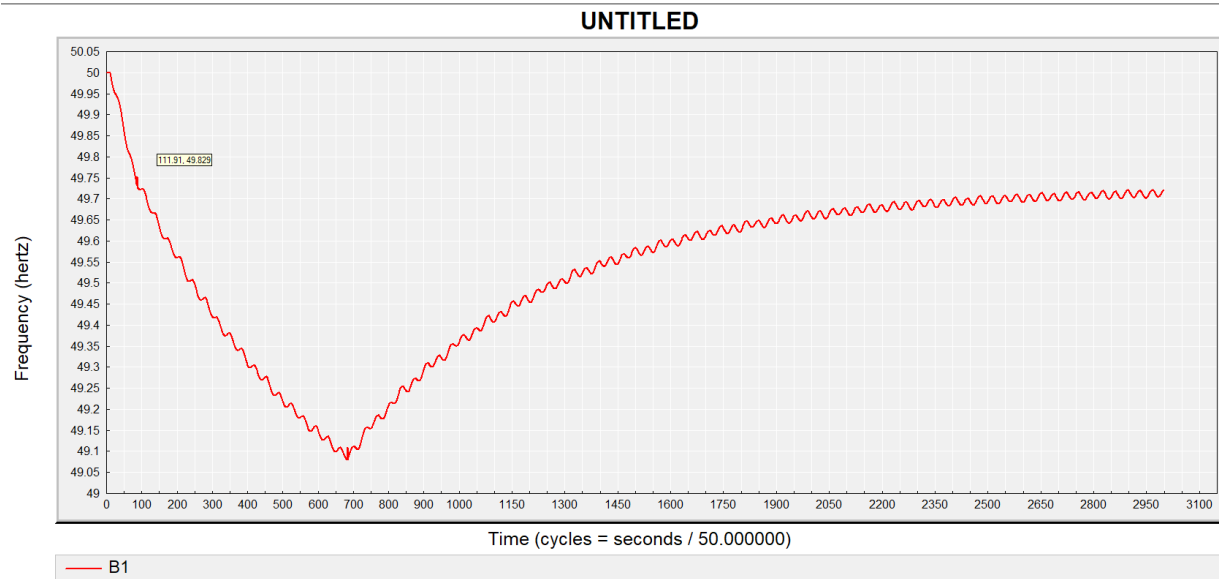
Help

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	35
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	40
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	50
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 49.075 Hz.

Zone 1: Case 4:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

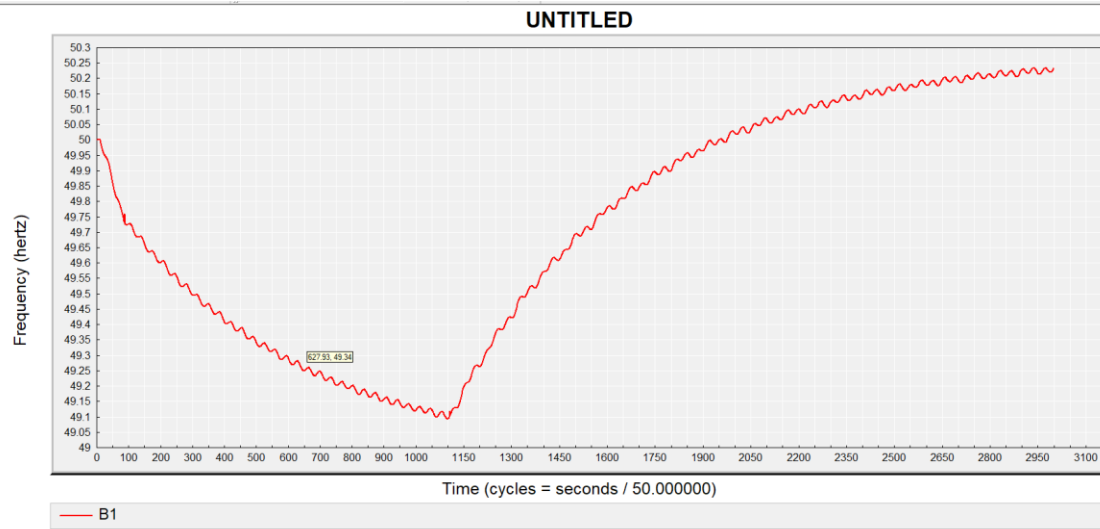
Database ID:

Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	45
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	50
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	60.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: 49.09 Hz.

Zone 1: Case 5:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

New

Delete

Reload

Default

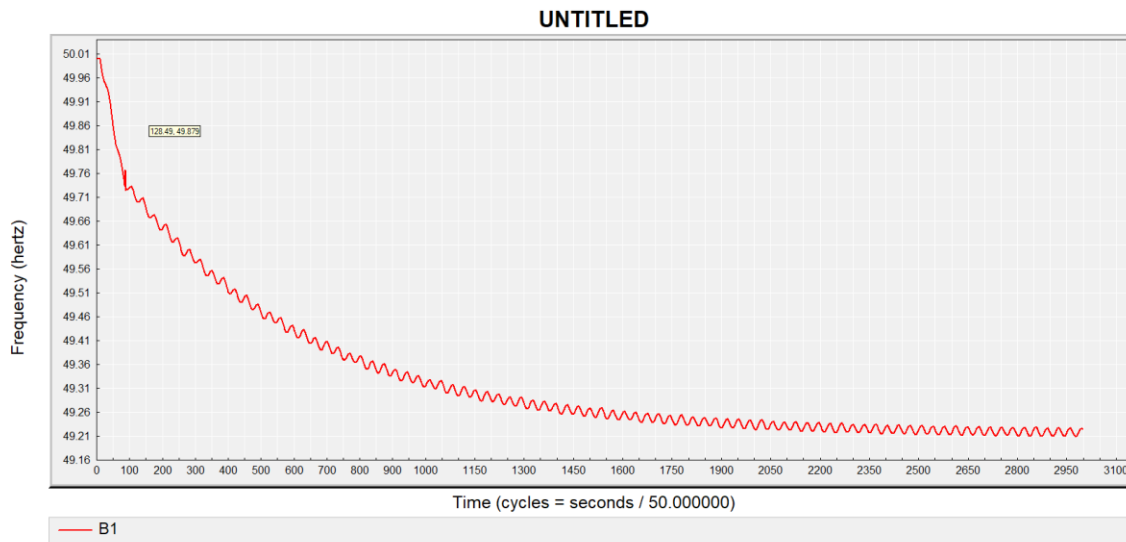
Help

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	55
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	60
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	70
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: Unstable so no nadir point found.

Analysis:

At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively. For zone 1, we analyzed the system for 5 different load shedding profiles. For the 1st case, the load shedding was less (15%,20%, and 30% for 1st,2nd and 3rd threshold respectively) and we found from the simulation that the system is not stable and there is no nadir point.

When we increased the shed amount for the 2nd case (25%,30% and 40%), we were able to make the system stable and the nadir point was 49.075 Hz. For 3rd case, we increased shed amount to 35%, 40% and 50% and we got the nadir point to be 49.075 Hz. For the 4th case, we again increased the shed amount to 45%,50% and 60% and got the nadir to be increased a bit to 49.09 Hz.

But when we further increase the shed amount for the 5th case to 55%,60% and 70%, we found that the system became unstable and there was no nadir point.

In comparison with the previous Kp value to be 1%, for the shedding profile to be 45%,50% and 60%, the nadir was 49.08Hz, but here in this case of Kp=2%, the nadir increased to 49.09Hz.

Zone 2: Case 1:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay
UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

43

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

47

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

58

Relay operating time [sec.]:

0.10000

New

Delete

Reload

Default

Help

EQ / TOTAL

1 / 1

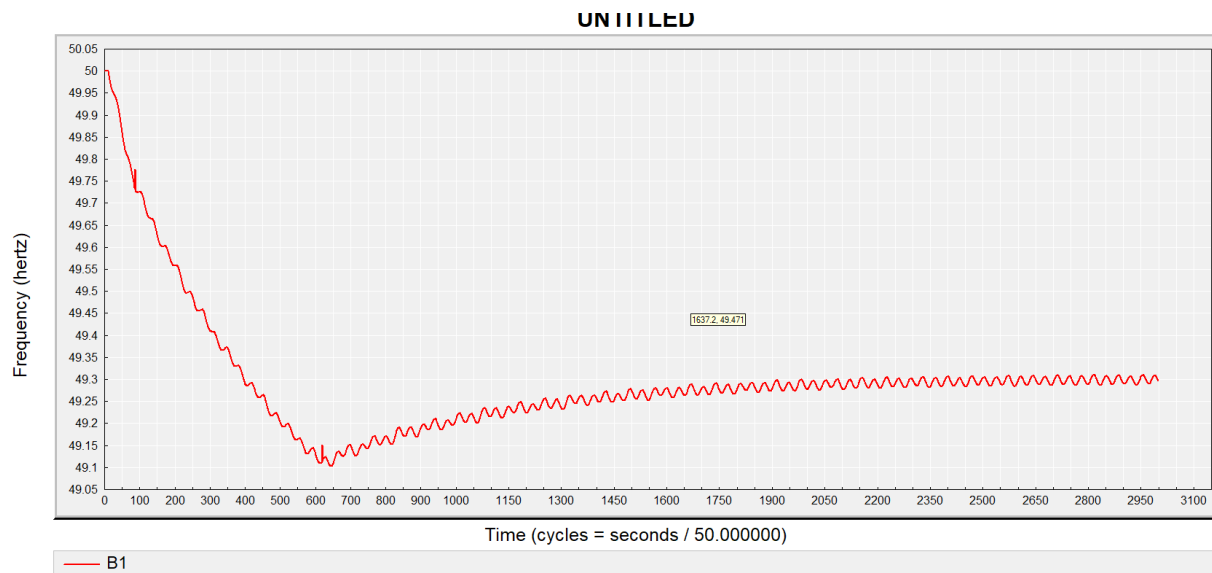
OK

CANCEL

<<

>>

Simulation:



Nadir Point: 49.1 Hz.

Zone 2: Case 2:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

First frequency threshold [Hz]: 49.80000

First load shedding amount [%]: 53

Second frequency threshold [Hz]: 49.10000

Second load shedding amount [%]: 57

Third frequency threshold [Hz]: 48.60000

Third load shedding amount [%]: 68

Relay operating time [sec.]: 0.10000

New

Delete

Reload

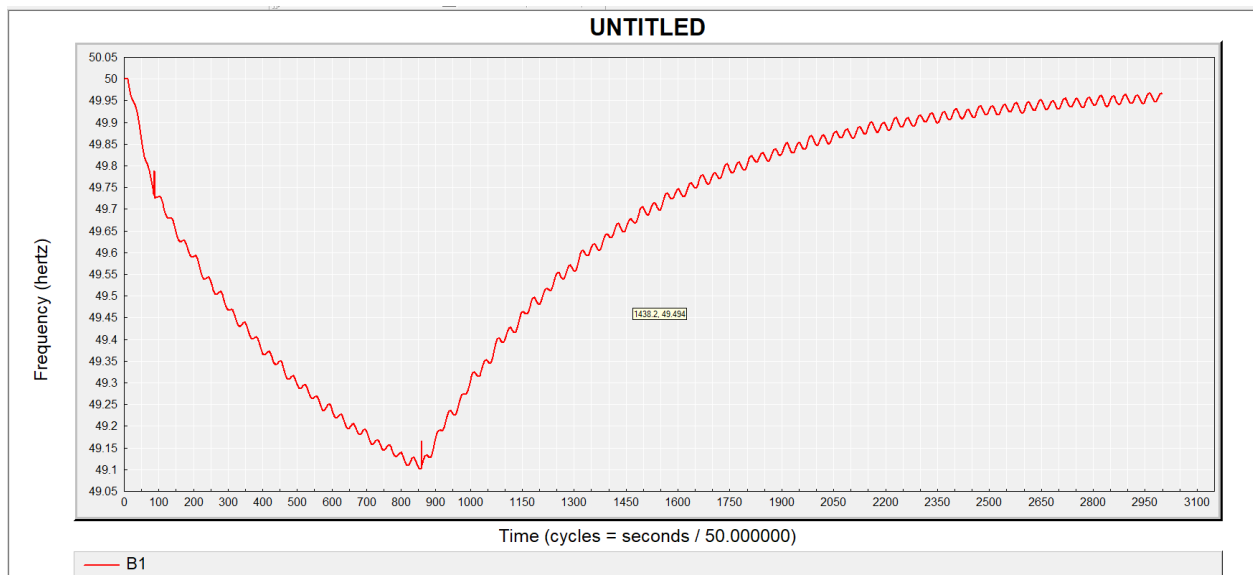
Default

Help

EQ / TOTAL
1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 49.1 Hz.

Zone 2: Case 3:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

Database ID:

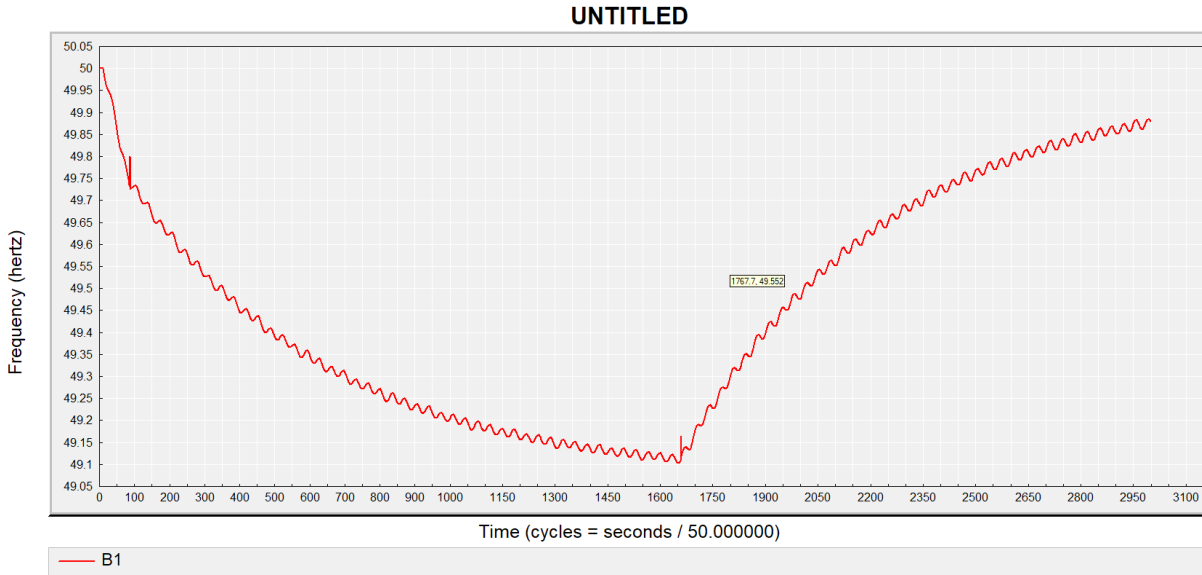
Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	63.00000
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	67.00000
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	78.00000
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

<<"/>

Simulation:



Nadir Point: 49.1 Hz.

Analysis:

At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively for all the cases. For zone 2, we analyzed the system for 3 different load shedding profiles. For the 1st case, the load shedding was less (43%, 47%, and 58% for 1st, 2nd and 3rd threshold respectively) and we found from the simulation that the system is stable and the nadir point to be 49.1 Hz.

When we increased the shed amount for the 2nd case (53%, 57% and 68%) and 3rd case (63%, 67% and 78%), the system was still stable and the nadir point was the same for the both cases (49.1 Hz).

From our experience we could say that, if the shed amount was very less, the system's stability should have been damaged.

In comparison with the previous Kp value to be 1%, for the shedding profile to be 43%, 47% and 58%, the nadir was 49.08Hz, but here in this case of Kp=2%, the nadir increased to 49.1Hz.

Zone 3: Case 1:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

20

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

30

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

40

Relay operating time [sec.]:

0.10000

EQ / TOTAL

1 / 1

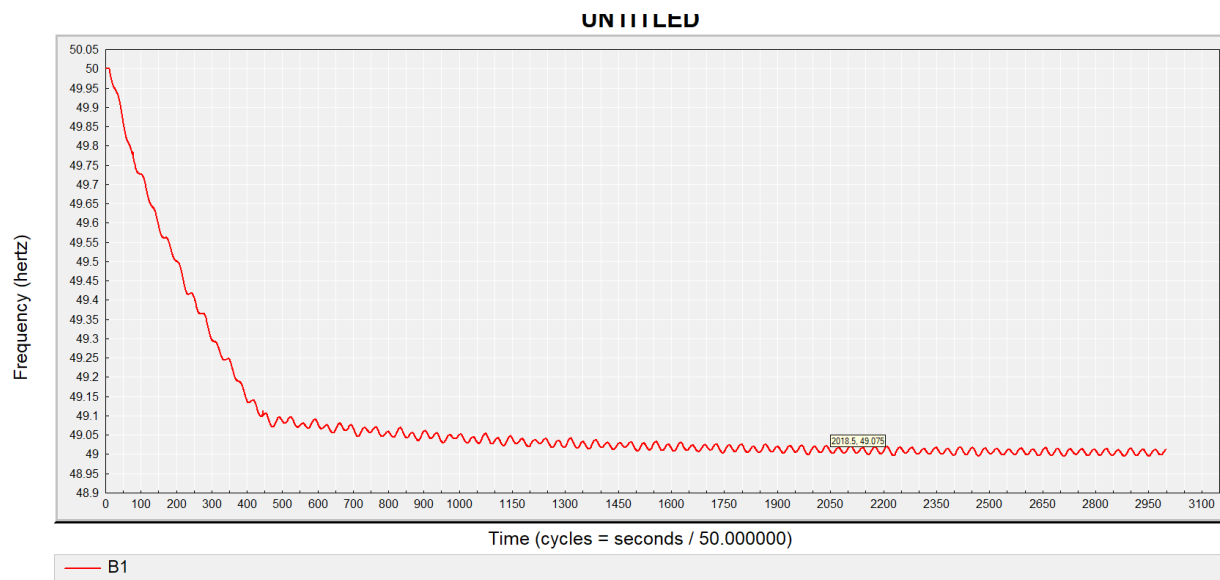
OK

CANCEL

<<

>>

Simulation:



Nadir Point: Unstable so no nadir point found.

Zone 3: Case 2:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

Database ID:

Connected to bus:

First frequency threshold [Hz]:

First load shedding amount [%]:

Second frequency threshold [Hz]:

Second load shedding amount [%]:

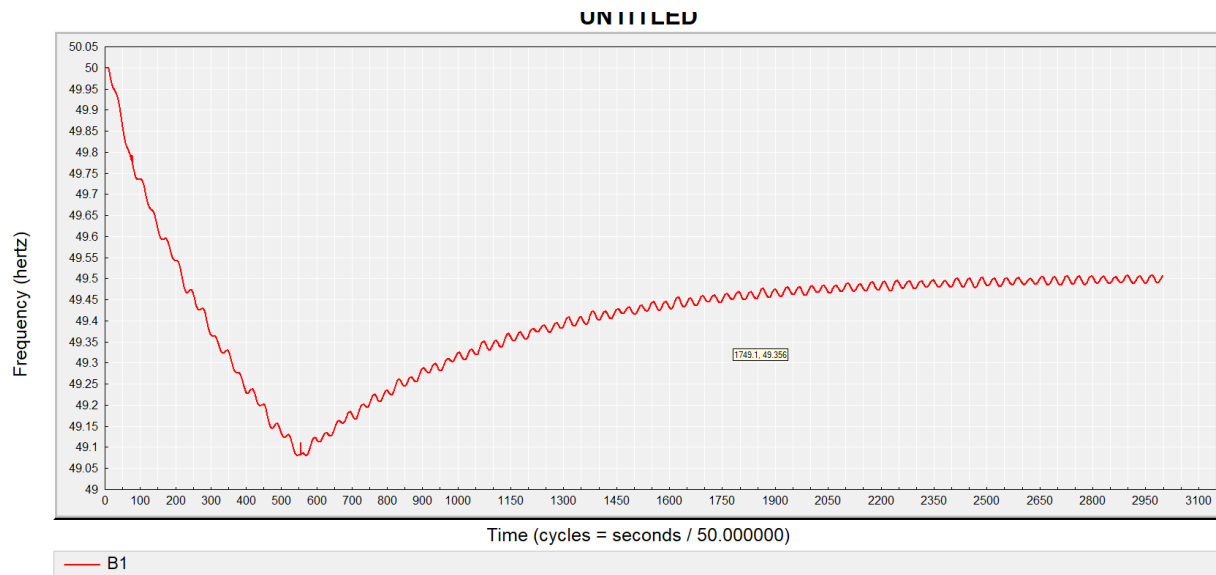
Third frequency threshold [Hz]:

Third load shedding amount [%]:

Relay operating time [sec.]:

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: 49.075 Hz.

Zone 3: Case 3:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES

Access DB

Connected to bus: GlobalRelay

Make Available To

First frequency threshold [Hz]: 49.80000

First load shedding amount [%]: 40

Second frequency threshold [Hz]: 49.10000

Second load shedding amount [%]: 50

Third frequency threshold [Hz]: 48.60000

Third load shedding amount [%]: 60

Relay operating time [sec.]: 0.10000

New

Delete

Reload

Default

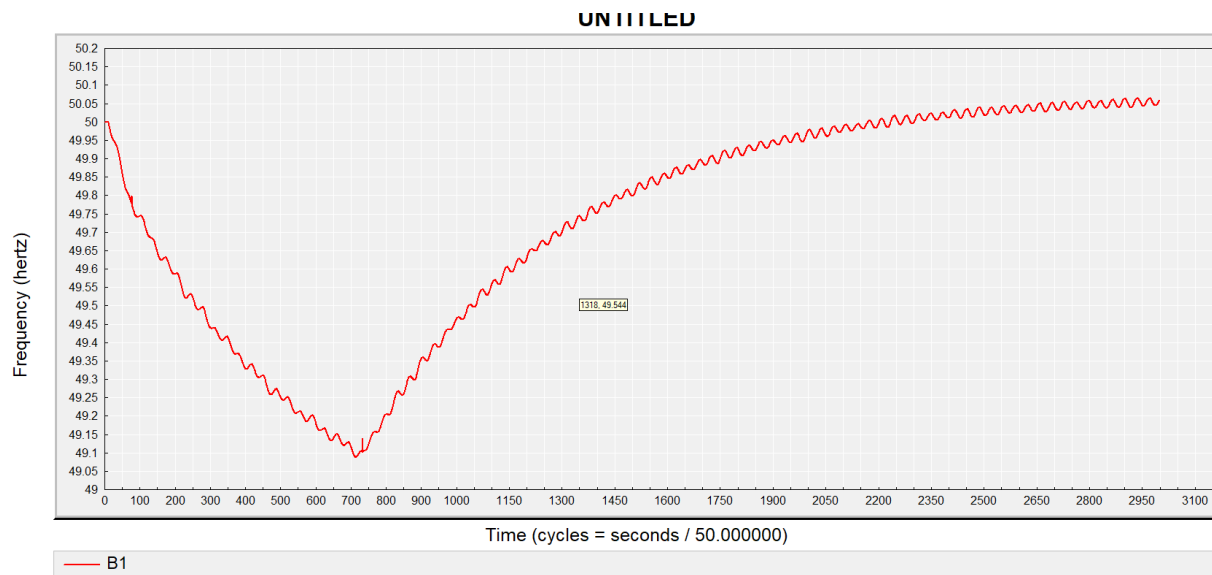
Help

EQ / TOTAL

1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 49.085 Hz.

Analysis:

At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively. For zone 3, we analyzed the system for 3 different load shedding profiles. For the 1st case, the load shedding was taken less (20%,30%, and 40% for 1st,2nd and 3rd threshold respectively) as this zone is the strongest one and we found from the simulation that the system is not stable and there is no nadir point.

When we increased the shed amount for the 2nd case (30%,40% and 50%), we were able to make the system stable and the nadir point was 49.075 Hz. For 3rd case, we increased shed amount to 40%,50% and 60% and we got the nadir point to be increased to 49.085 Hz.

In comparison with the previous Kp value to be 1%, for the shedding profile to be 20%,30% and 40%, the nadir was 48.59Hz, but here in this case of Kp=2%, for the same profile the system was unstable, but increasing the load shedding a bit to 30%,40% and 50%, we found the increased nadir to be 49.075Hz.

For Kp = 3% :

Transient Stability Study Dialog

Events	Monitoring	Real time plotting	Tabular report	Relay/Zones/Areas	Display Options
Title	Settings	LF data	Global Load	Load/Zone	Frequency/Zone
				Global Selection	UDM

Zload: 0.500 Voltage below which load is represented by const Z

nP: 2.0000 Exponent of voltage term in active load function

nQ: 2.0000 Exponent of voltage term in reactive load function

Pfreq: 3.00000 Correction factor in function: Load MW vs Frequency

Qfreq: 0.00000 Correction factor in function: Load MVAR vs Frequency

Default

Reload

OK CANCEL APPLY

Zone 0: Case 1:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay

UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

20

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

30

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

40

Relay operating time [sec.]:

0.10000

EQ / TOTAL

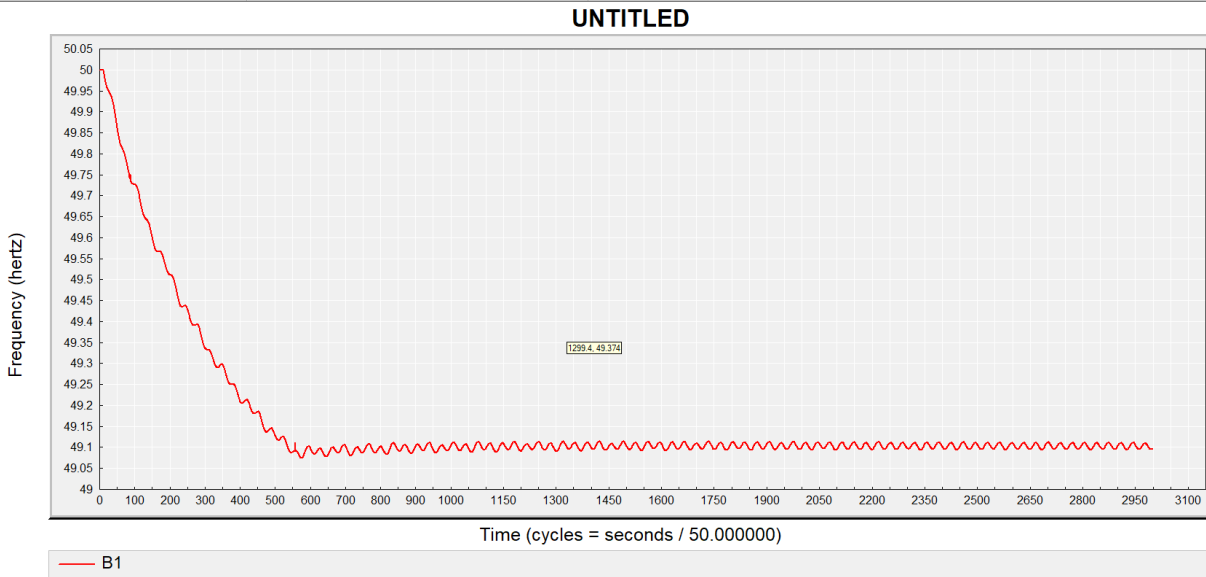
1 / 1

OK

CANCEL

<<

>>



Nadir Point: 49.07 Hz.

Zone 0: Case 2:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM |

Low Freq. ID:

Database ID:

Connected to bus:

First frequency threshold [Hz]:

First load shedding amount [%]:

Second frequency threshold [Hz]:

Second load shedding amount [%]:

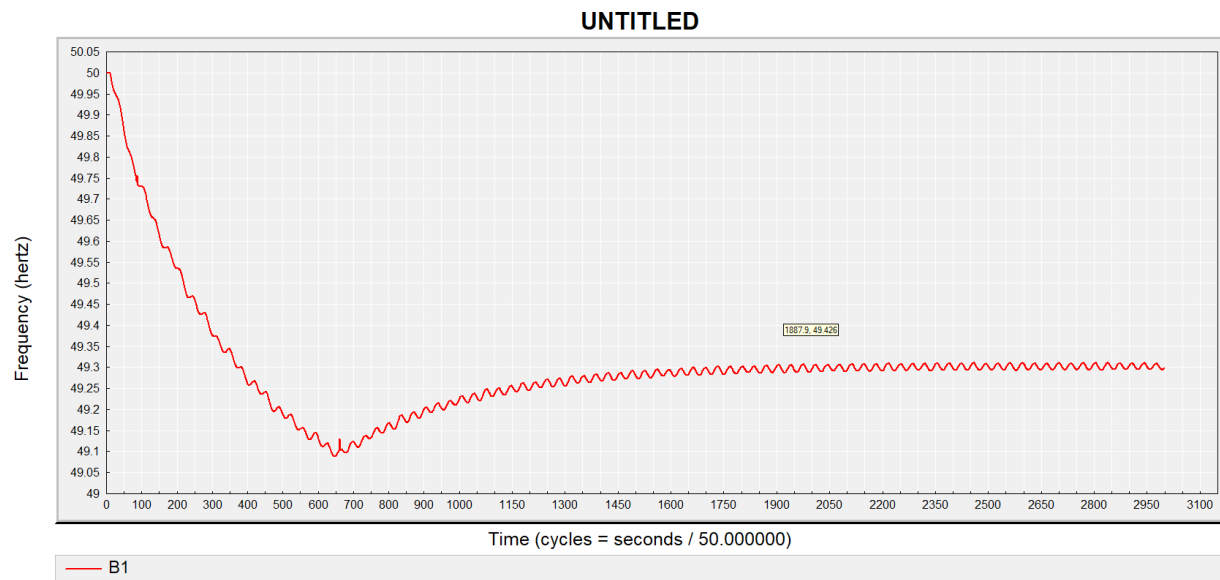
Third frequency threshold [Hz]:

Third load shedding amount [%]:

Relay operating time [sec.]:

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: 49.08 Hz.

Zone 0: Case 3:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

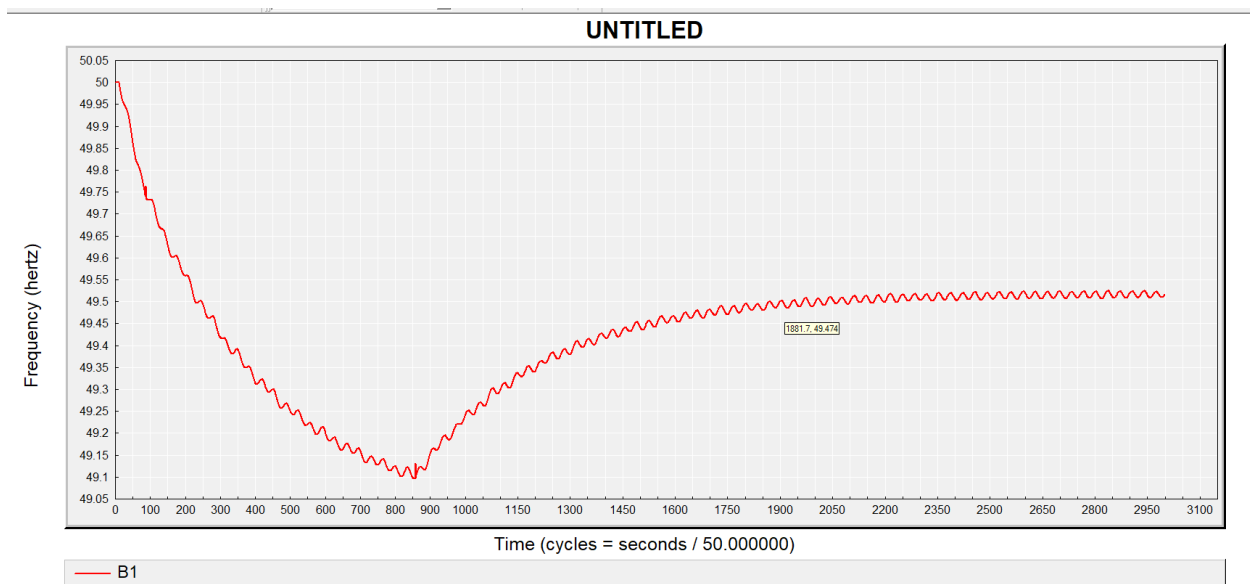
Database ID:

Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	40
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	50
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	60
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: 49.1 Hz.

Zone 0: Case 4:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

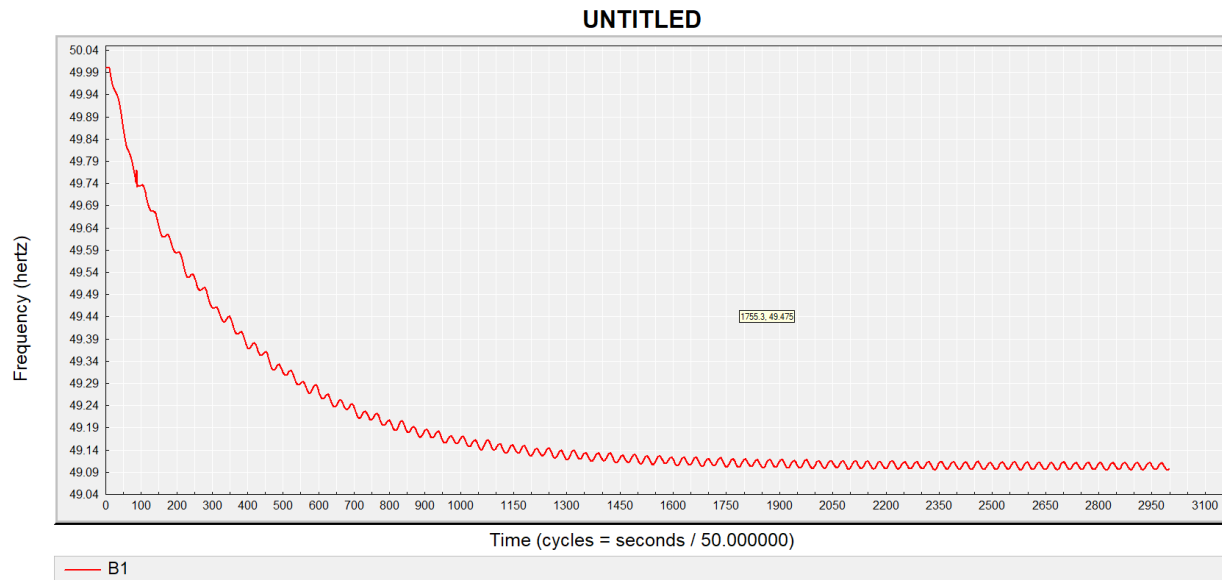
Database ID:

Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	50
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	60
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	70
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: Unstable so no nadir point found.

Zone 0: Case 5:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM |

Low Freq. ID:

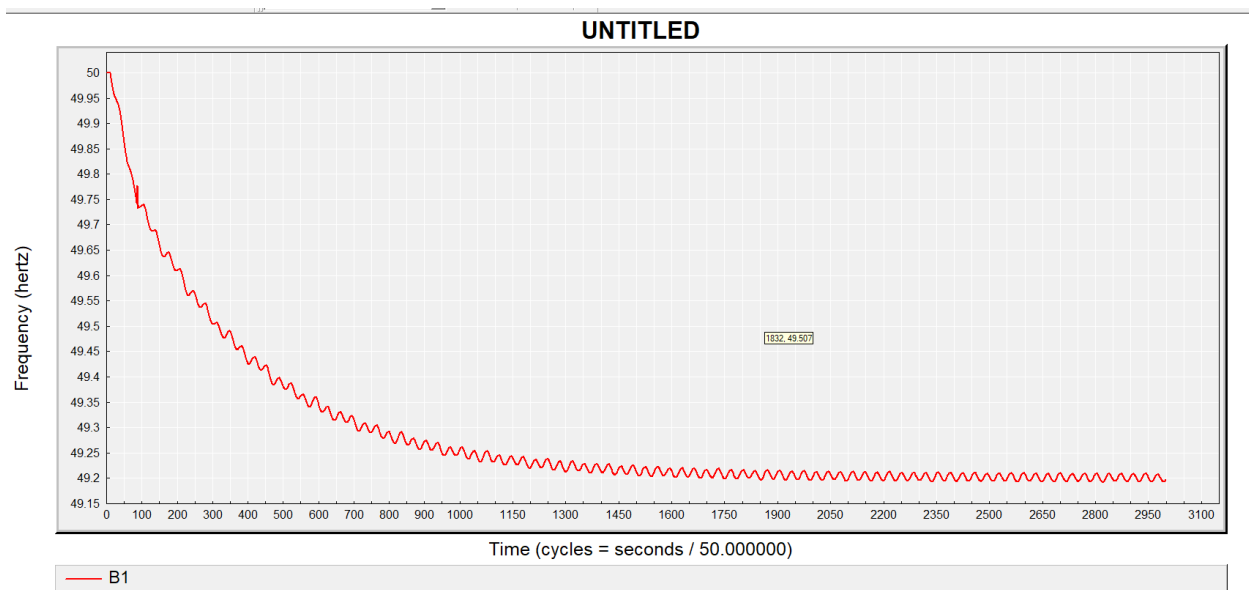
Database ID:

Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	60
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	70
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	80
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: Unstable so no nadir point found.

Zone 0: Case 6:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

New

Delete

Reload

Default

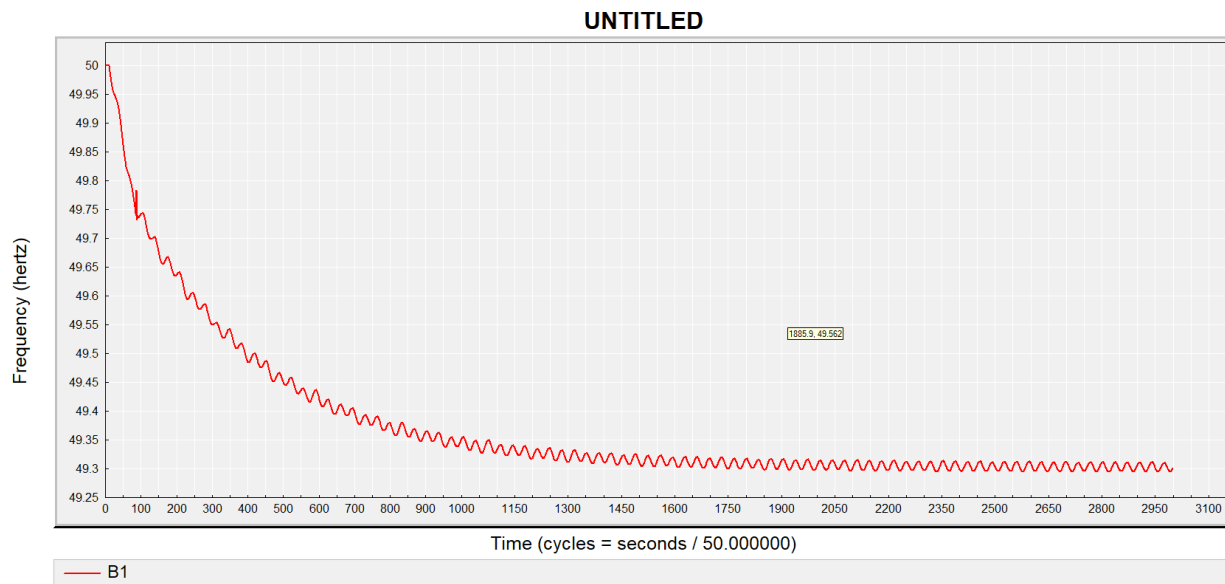
Help

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	70
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	80
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	90
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: Unstable so no nadir point found.

Analysis:

At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively. For zone 0, we analyzed the system for 6 different load shedding profiles. For the 1st case, the load shedding was taken less (20%,30%, and 40% for 1st,2nd and 3rd threshold respectively) and we found from the simulation that the nadir point to be 49.07.

When we increased the shed amount for the 2nd case (30%,40% and 50%), the nadir point was increased a bit to 49.08 Hz and for the 3rd case, we further increased shed amount to 40%,50% and 60% and we got the nadir point to be increased to 49.1 Hz.

But when we again increased the shed amount for the next cases (for 4th case 50%,60%,70%, for 5th case 60%,70%,80%, for 6th case 70%,80%,90%), every time we found the system to be unstable and no nadir point was detected. That means, we were able to increase the shed amount upto certain value as well as could improve the nadir point but further increasement made the system unstable.

In comparison with the previous Kp value to be 2%, for the shedding profile to be 30%,40% and 50%, the system was unstable, but here in this case of Kp=3%, for the same profile the nadir was 49.08Hz. Increasing the load shedding a bit to 40%,50% and 60%, we found the nadir to be 49.075Hz for Kp=2%, but here for Kp=3 %, the nadir increased to 49.1 Hz. Again, for the increased profile of 50%,60% and 70%, the nadir was 49.09Hz for Kp=2%, but for the same profile of Kp=3%, the system was unstable.

Zone 1: Case 1:

Load Shedding profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

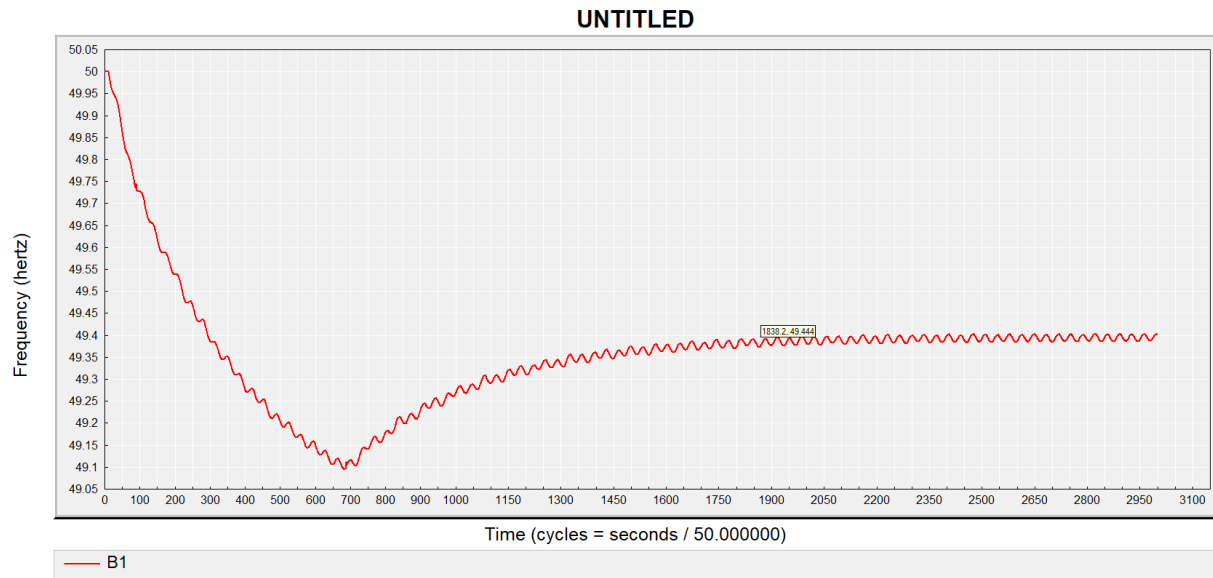
New
Delete
Reload
Default
Help

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	20
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	30
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	40
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 49.09 Hz.

Zone 1: Case 2:

Load Shedding profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

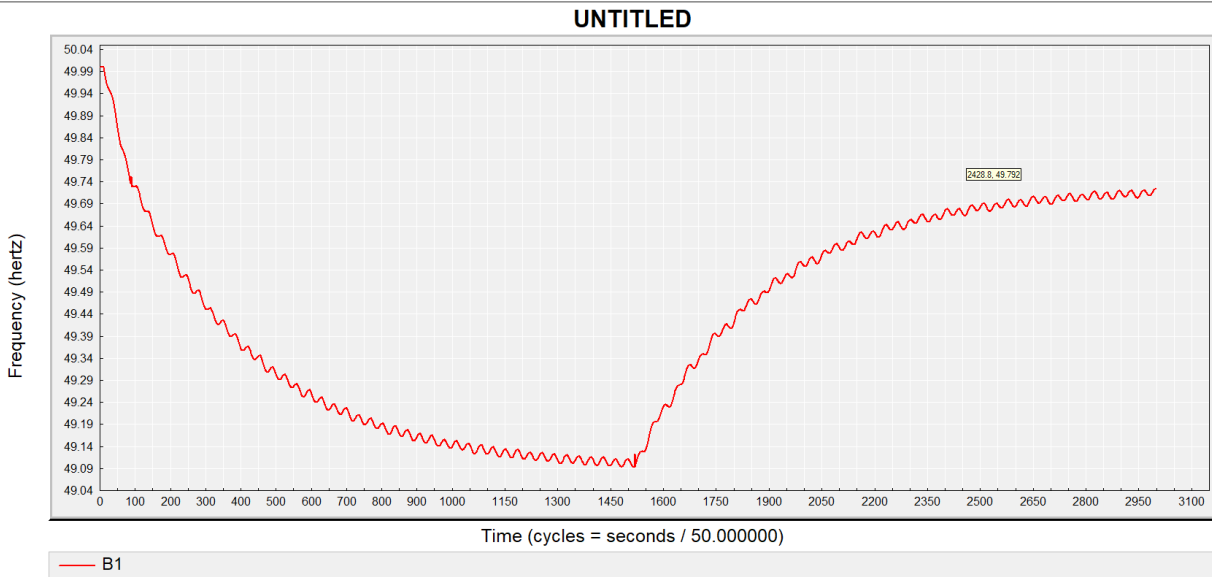
Database ID:

Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	30
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	40
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	50
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

Simulation:



Nadir Point: 49.09 Hz.

Zone 1: Case 3:

Load Shedding profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID:

Database ID:

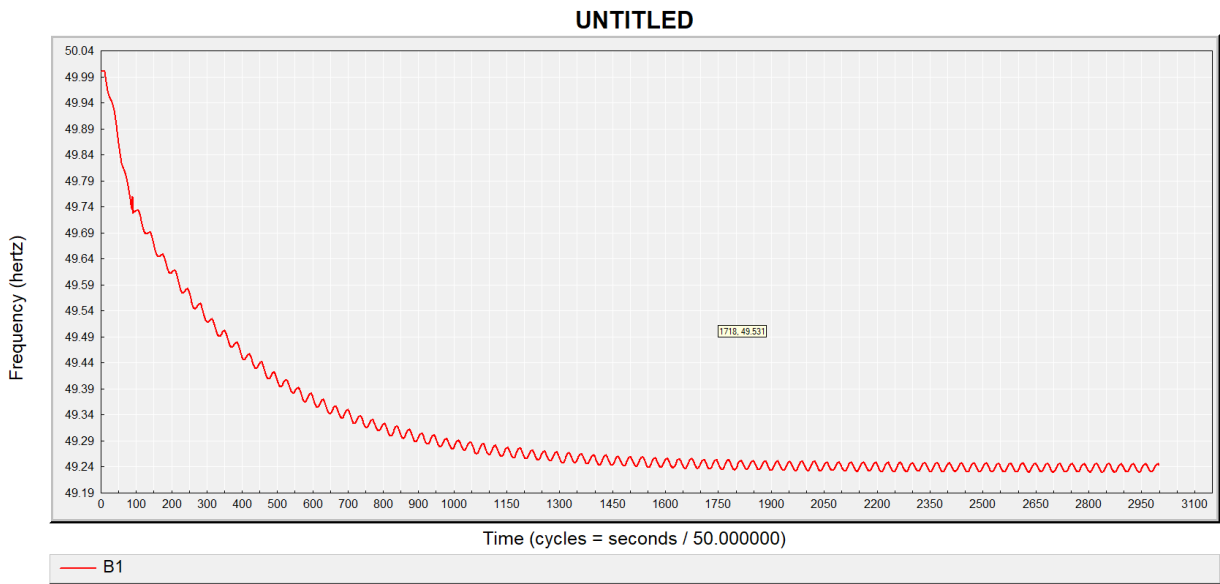
Connected to bus:

First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	40
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	50
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	60
Relay operating time [sec.]:	0.10000

EQ / TOTAL
1 / 1

<< >>

Simulation:



Nadir Point: Unstable so no nadir point found.

Zone 1: Case 4:

Load Shedding profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

New

Delete

Reload

Default

Help

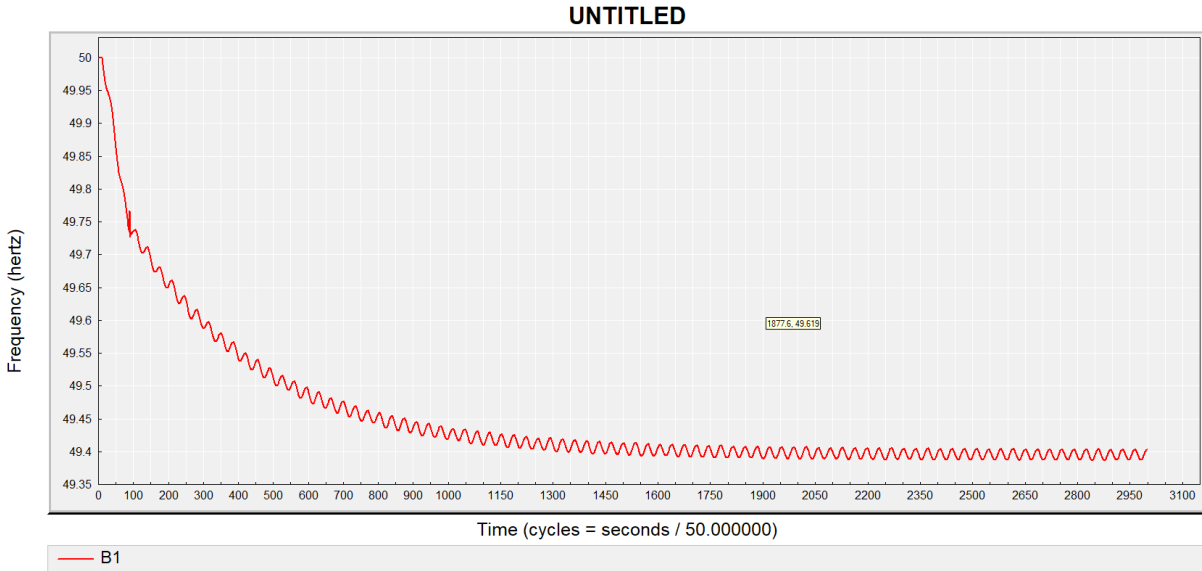
First frequency threshold [Hz]:	49.80000
First load shedding amount [%]:	50
Second frequency threshold [Hz]:	49.10000
Second load shedding amount [%]:	60
Third frequency threshold [Hz]:	48.60000
Third load shedding amount [%]:	70
Relay operating time [sec.]:	0.10000

EQ / TOTAL

1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: Unstable so no nadir point found.

Analysis:

At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively. For zone 1, we analyzed the system for 4 different load shedding profiles. For the 1st case, the load shedding was taken less (20%,30%, and 40% for 1st,2nd and 3rd threshold respectively) and we found from the simulation that the nadir point to be 49.09.

When we increased the shed amount for the 2nd case (30%,40% and 50%), the nadir point remains same as 49.09 Hz. For the 3rd case, we further increased shed amount to 40%,50% and 60% and we found the system unstable. We increased the shed amount again (50%,60% and 70%) and still the system was unstable.

That means the system was stable upto case 2 and further increasement in shed amount made the system unstable.

In comparison with the previous Kp value to be 2%, for the shedding profile to be 35%,40% and 50%, the nadir point was 49.075Hz, but here in this case of Kp=3% for the shedding profile to be 30%,40% and 50%, the nadir was increased to 49.09Hz. Increasing the load shedding a bit to 45%,50% and 60%, we found the nadir to be 49.09Hz for Kp=2%, but here for Kp=3%, for the shedding profile to be 40%,50% and 60%, the system became unstable. Again, for the increasement of load shedding profile, the system was unstable for both the Kp values to be 2% and 3%.

Zone 2: Case 1:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

First frequency threshold [Hz]: 49.80000

First load shedding amount [%]: 29

Second frequency threshold [Hz]: 49.10000

Second load shedding amount [%]: 38

Third frequency threshold [Hz]: 48.60000

Third load shedding amount [%]: 48

Relay operating time [sec.]: 0.10000

New

Delete

Reload

Default

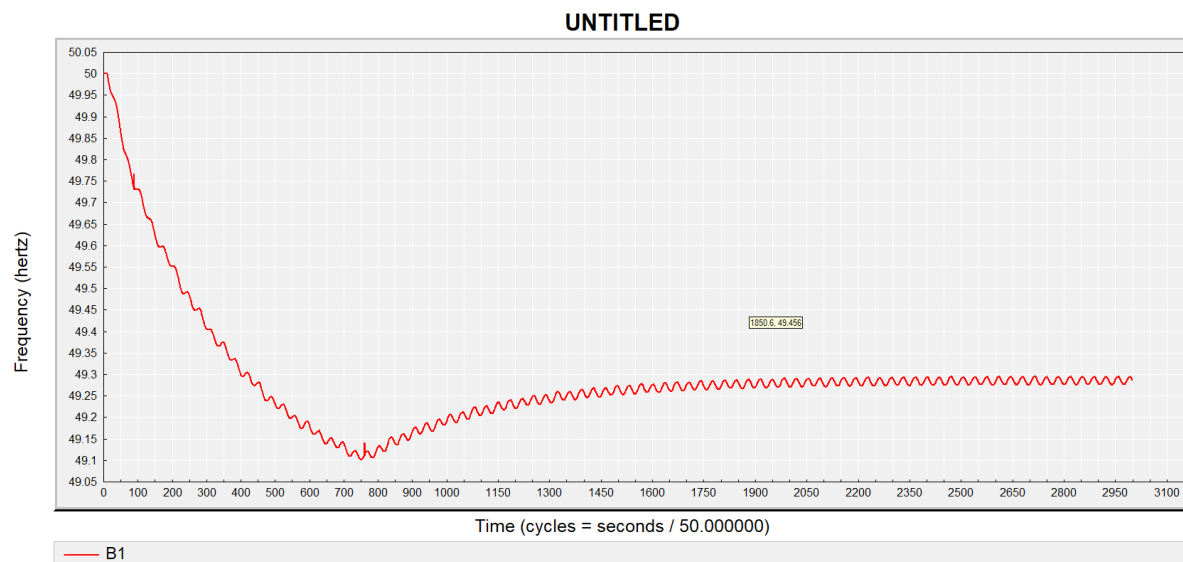
Help

EQ / TOTAL

1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 49.1 Hz.

Zone 2: Case 2:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

First frequency threshold [Hz]: 49.80000

First load shedding amount [%]: 39

Second frequency threshold [Hz]: 49.10000

Second load shedding amount [%]: 48

Third frequency threshold [Hz]: 48.60000

Third load shedding amount [%]: 58

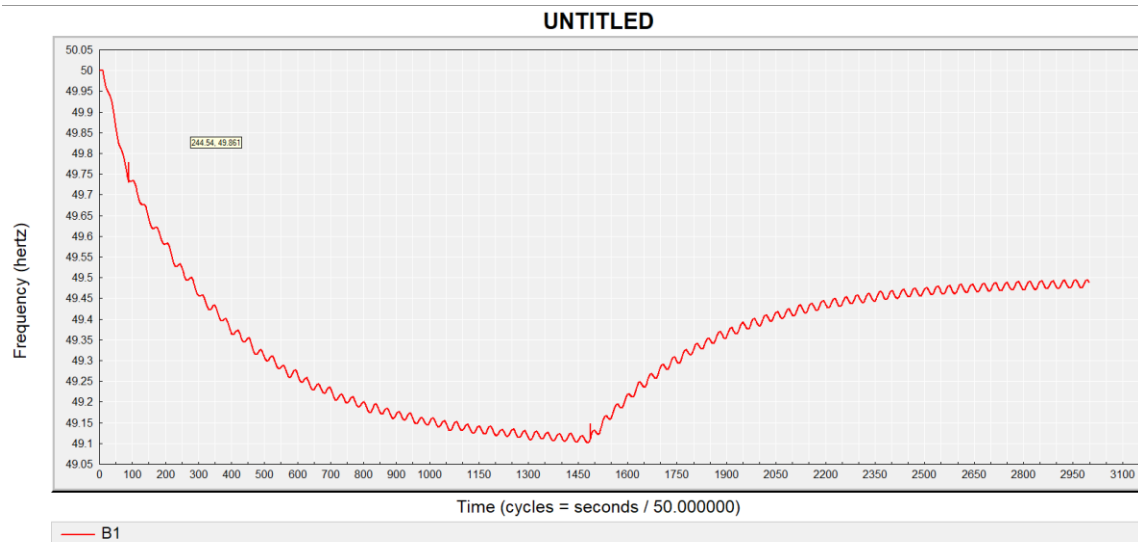
Relay operating time [sec.]: 0.10000

New Delete Reload Default Help

EQ / TOTAL 1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 49.1 Hz

Zone 2: Case 3:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay

UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

49

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

58

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

68

Relay operating time [sec.]:

0.10000

New

Delete

Reload

Default

Help

EQ / TOTAL

1 / 1

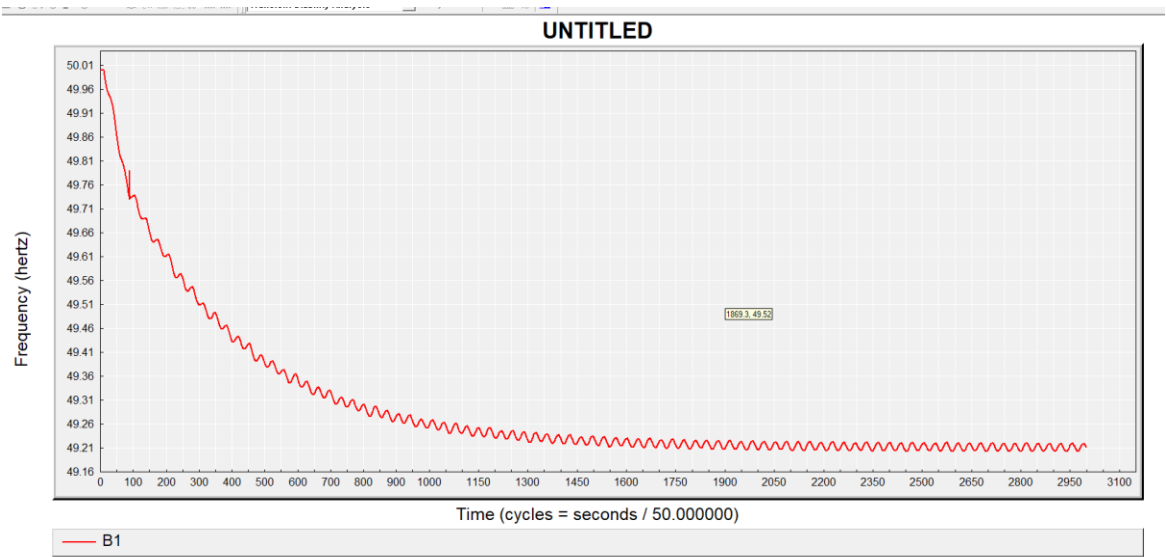
OK

CANCEL

<<

>>

Simulation:



Nadir Point: Unstable so no nadir point found.

Analysis:

At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively. For zone 2, we analyzed the system for 3 different load shedding profiles. For the 1st case, the load shedding was taken less (29%,38%, and 48% for 1st,2nd and 3rd threshold respectively) and we found from the simulation that the nadir point to be 49.1 Hz.

When we increased the shed amount for the 2nd case (39%,48% and 58%), the nadir point remains same as 49.1 Hz. For the 3rd case, we further increased shed amount to 49%,58% and 68% and we found the system unstable.

That means the system was stable upto case 2 and further increasement in shed amount made the system unstable.

In comparison with the previous Kp value to be 2%, for the shedding profile to be 43%,47% and 58%, the nadir point was 49.1Hz, but here in this case of Kp=3% for the shedding profile to be 39%,48% and 58%, the nadir was exactly same 49.1Hz. Increasing the load shedding a bit to 53%,57% and 68%, we found the nadir to be 49.1Hz for Kp=2%, but here for Kp=3%, for the shedding profile to be 49%,58% and 68%, the system became unstable. Again, for the increasement of load shedding profile, the system was unstable for Kp=3%, but the system was stable for Kp= 2%(nadir was 49.1Hz).

Zone 3: Case 1:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

10

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

20

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

30

Relay operating time [sec.]:

0.10000

New

Delete

Reload

Default

Help

EQ / TOTAL

1 / 1

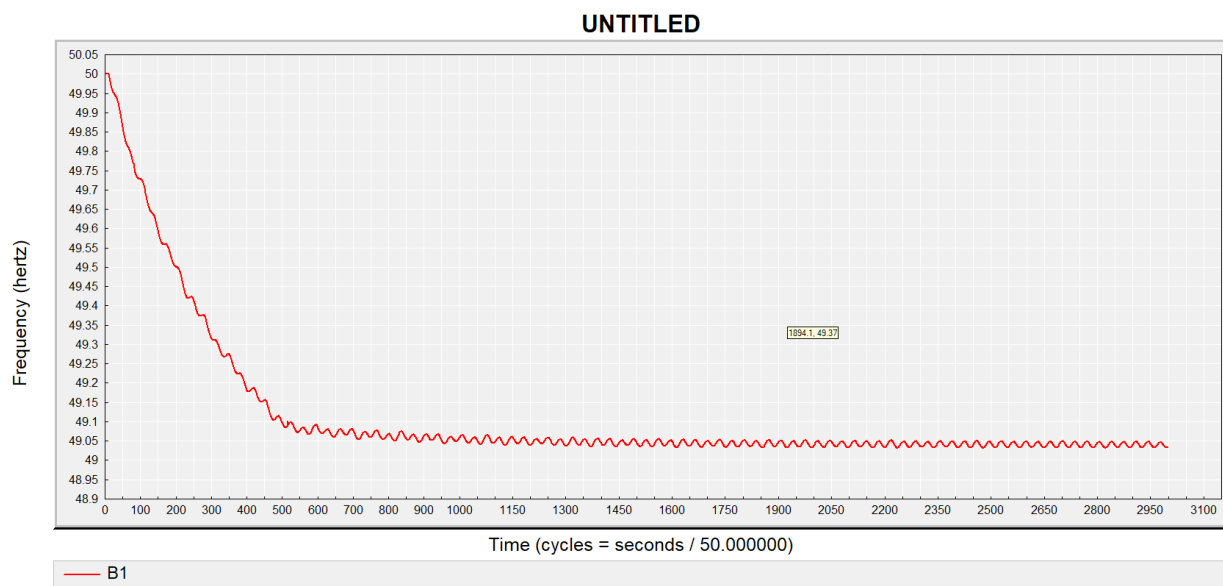
OK

CANCEL

<<

>>

Simulation:



Nadir Point: Unstable so no nadir point found.

Zone 3: Case 2:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES

Access DB

Connected to bus: GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]: 49.80000

First load shedding amount [%]: 20

Second frequency threshold [Hz]: 49.10000

Second load shedding amount [%]: 30

Third frequency threshold [Hz]: 48.60000

Third load shedding amount [%]: 40

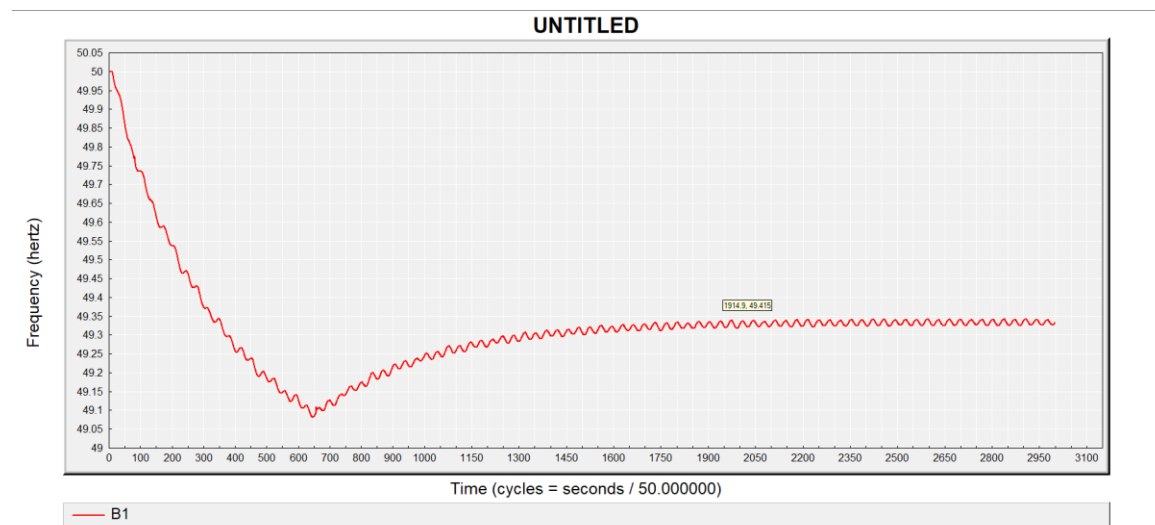
Relay operating time [sec.]: 0.10000

EQ / TOTAL

1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: 49.08 Hz.

Zone 3: Case 3:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay

UDM

Low Freq. ID

NEW ITEM

Database ID

DEF_VALUES

Access DB

Connected to bus:

GlobalRelay

Make Available To

New

Delete

Reload

Default

Help

First frequency threshold [Hz]:

49.80000

First load shedding amount [%]:

30

Second frequency threshold [Hz]:

49.10000

Second load shedding amount [%]:

40

Third frequency threshold [Hz]:

48.60000

Third load shedding amount [%]:

50

Relay operating time [sec.]:

0.10000

EQ / TOTAL

1 / 1

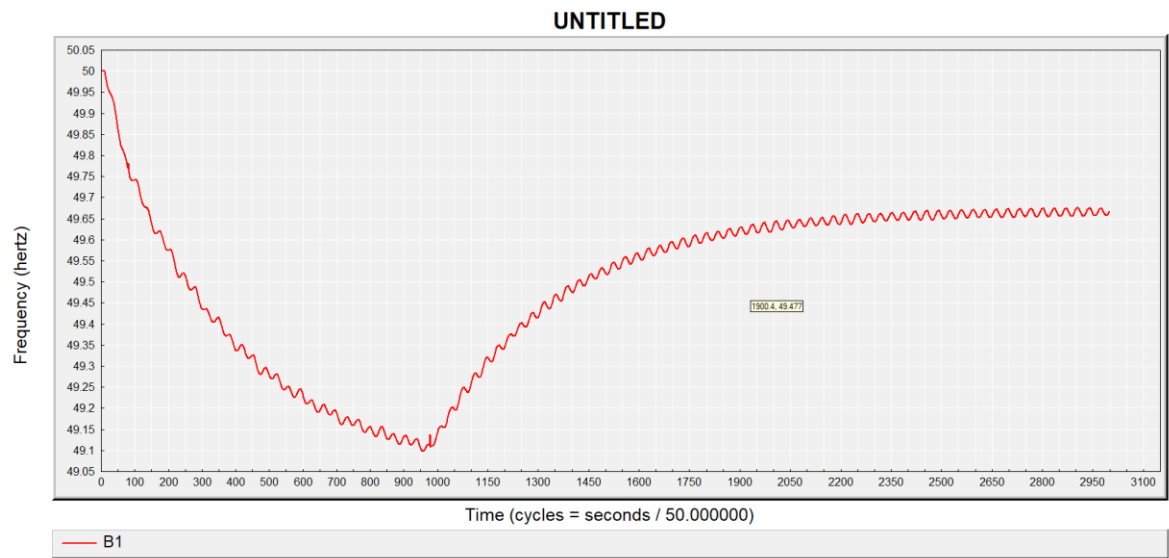
OK

CANCEL

<<

>>

Simulation:



Nadir Point: 49.095 Hz.

Zone 3: Case 4:

Load Shedding Profile:

Global Low Frequency Relay

Low Frequency Relay | UDM

Low Freq. ID: NEW ITEM

Database ID: DEF_VALUES Access DB

Connected to bus: GlobalRelay Make Available To

First frequency threshold [Hz]: 49.80000

First load shedding amount [%]: 40

Second frequency threshold [Hz]: 49.10000

Second load shedding amount [%]: 50

Third frequency threshold [Hz]: 48.60000

Third load shedding amount [%]: 60

Relay operating time [sec.]: 0.10000

New

Delete

Reload

Default

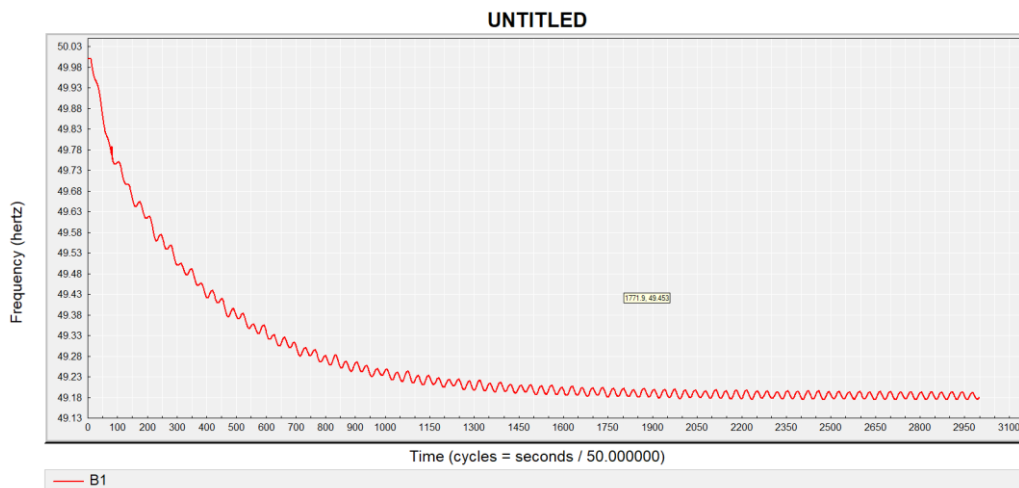
Help

EQ / TOTAL

1 / 1

OK CANCEL << >>

Simulation:



Nadir Point: Unstable so no nadir point found.

Analysis:

At first, we fixed the frequency threshold to be 49.8Hz, 49.1Hz and 48.6Hz for the 1st, 2nd and 3rd step respectively. For zone 3, we analyzed the system for 4 different load shedding profiles. For the 1st case, the load shedding was taken less (10%,20%, and 30% for 1st,2nd and 3rd threshold respectively) and we found from the simulation that the system was unstable.

When we increased the shed amount for the 2nd case (20%,30% and 40%), the system becomes stable and the nadir point was 49.08 Hz. For the 3rd case, we further increased shed amount to 30%,40% and 50% and we found the nadir point increased to 49.095 Hz.

For the 4th case we again increased the shed amount to 40%,50%,60% and got the system unstable. That means the system was stable for the 2nd and 3rd cases but less than a certain amount of shedding and also greater than a certain amount of shedding of loads made the system unstable.

In comparison with the previous Kp value to be 2%, for the shedding profile to be 20%,30% and 40%, the system was unstable, but here in this case of Kp=3% for the same shedding profile, the system is stable and the nadir was 49.08Hz. Increasing the load shedding to 30%,40% and 50%, we found the nadir to be 49.075Hz for Kp=2%, but here for Kp=3%, for the same shedding profile, the nadir point was increased to 49.095Hz. Again, for the increase of load shedding profile to 40%,50% and 60%, the nadir point was 49.085Hz for Kp=2%, but for the same profile, the system became unstable for Kp=3%.

Conclusion

In this project, we extensively studied the critical aspect of frequency stability in power systems, focusing on a 39-bus network with various components including 34 lines, 12 transformers, 10 generators, and 19 loads. Our primary aim was to analyze the system's response to generation loss events, with particular emphasis on the role of synchronous generators in damping frequency declines. By conducting a thorough frequency stability study, we highlighted the importance of maintaining consistent frequency levels post-major disturbances and explored the efficacy of Under Frequency Load Shedding (UFLS) mechanisms as a backup solution to stabilize the system. Our analyses confirmed that synchronous generators, with their inherent mechanical inertia, play a vital role in mitigating initial frequency deviations, thereby providing a buffer for subsequent corrective actions.

We employed several analytical methods, including power flow analysis and transient stability analysis, to assess the power system's performance under various conditions. Power flow analysis enabled us to determine the voltage, current, power, and phase angle at different points in the system under steady-state conditions. Meanwhile, transient stability analysis provided insights into the system's dynamic response to disturbances, revealing critical aspects of its behavior over short periods. Through this comprehensive approach, we evaluated frequency excursions following generation loss and assessed the effectiveness of load shedding schemes in preventing catastrophic failures and maintaining overall system stability.

Furthermore, we calculated the Fast Voltage Stability Index (FVSI) to identify potential voltage instability points and divided the network into zones, distinguishing stronger and weaker areas. We developed several load shedding profiles, simulated various scenarios, and analyzed the resulting frequency response curves, particularly focusing on the system frequency or Nadir point. By optimizing the load shedding profiles, we were able to enhance the system's resilience to disturbances. Our findings underscore the importance of strategic load shedding and robust frequency control measures in ensuring a reliable and resilient power grid, capable of withstanding major contingencies without compromising stability.

Our Individual Contributions:

Tahsin Jawad Khan- Input Parameter Calculation, Data Input, Frequency Response Analysis, Preparing Presentation.

Muhid Rahman- Input Parameter Calculation, Data Input, Load Flow Analysis, Preparing Presentation.

Shafayet Sadik Sowad- FVSI calculation, Zone sectorization and Zone Optimization by Simulation, Frequency Response Analysis, Preparing Presentation.

K M Istiaque- Software Simulation, Designing Load Shedding Scheme and Analysis, Preparing Presentation.

Sadman Shafi Ifty- Software Simulation, Designing Load Shedding Scheme and Analysis, Report writing.

THE END